

Introduction to MLOps

Vladislav Goncharenko



MIPT, fall 2026

Vladislav Goncharenko

- Author of machine learning courses and Masters program at MIPT
- Associate professor at MSU, HSE, Harbour.space
- ML researcher (MIPT)
- Team lead of video ranking team at Dzen (yandex.ru)
- Ex-team lead of perception team at self-driving trucks (Evocargo)
- Open source fan



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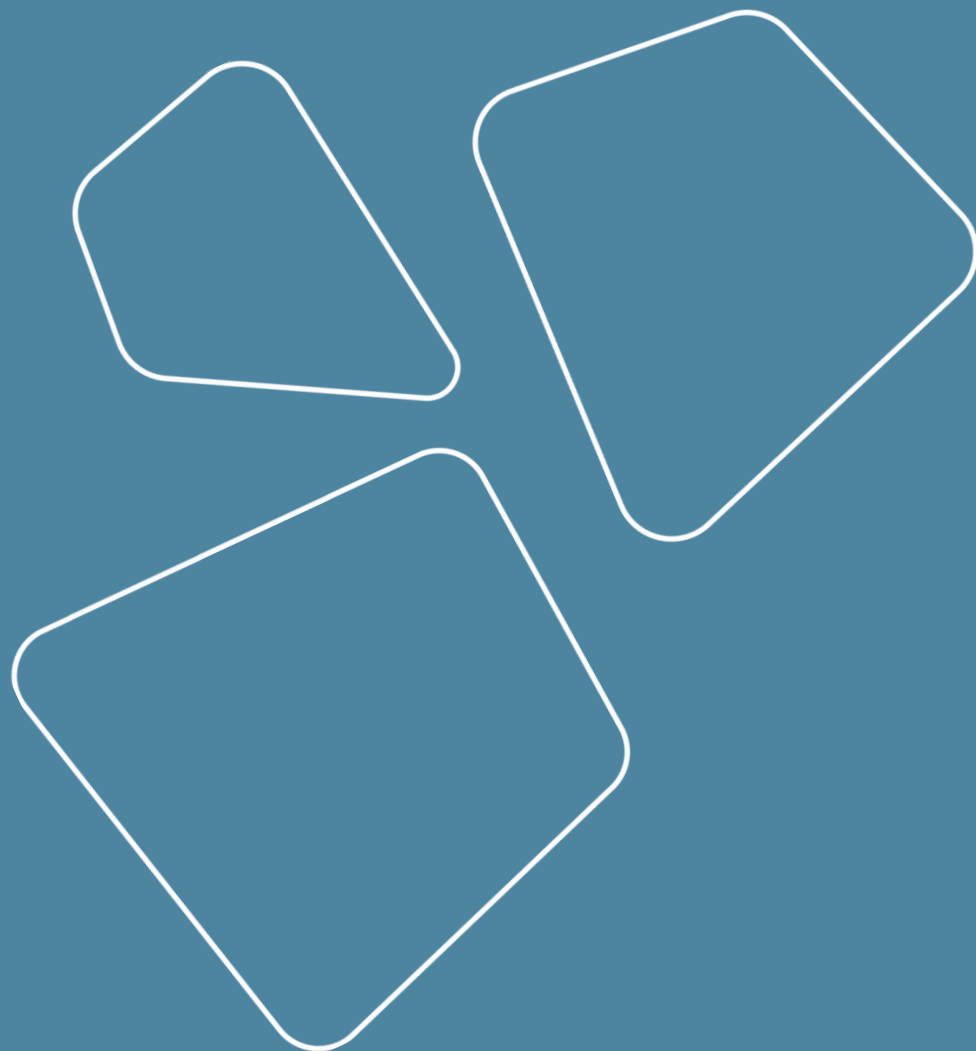
Дзен

Team

- Demid Gaibov is main seminarist responsible for practical part
- Vladislav Naumov is author of practical sections, seminarist and assistant
- Eugene Astafurov contributed a lot to materials and course structure

Outline

1. Prerequisites
2. Motivation
3. What is MLOps?
4. Subtasks
 - a. course structure
5. Stacks overview
6. Homeworks outline



Preface

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Prerequisites

- UNIX command line basics
 - git basics
- Python programming experience
 - web services is a plus
- Basic machine learning knowledge
 - sklearn
 - pytorch

Disclaimer

- We discuss broad way to do things
- However there are plenty of ways
- We prefer ways suitable for most ML engineers
 - meaning not very deep software engineering skills
- Python is main tool for this course
 - MLOps exists in other stacks too, but for ML it's mostly Python

Your part

1. Knowledge level of each student is different
2. We adjust materials based on your knowledge
3. Your knowledge is estimated by feedback

More you ask - more you get from this course!

I can give you a way but you need to make an effort to follow it

Motivation for MLOps

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01

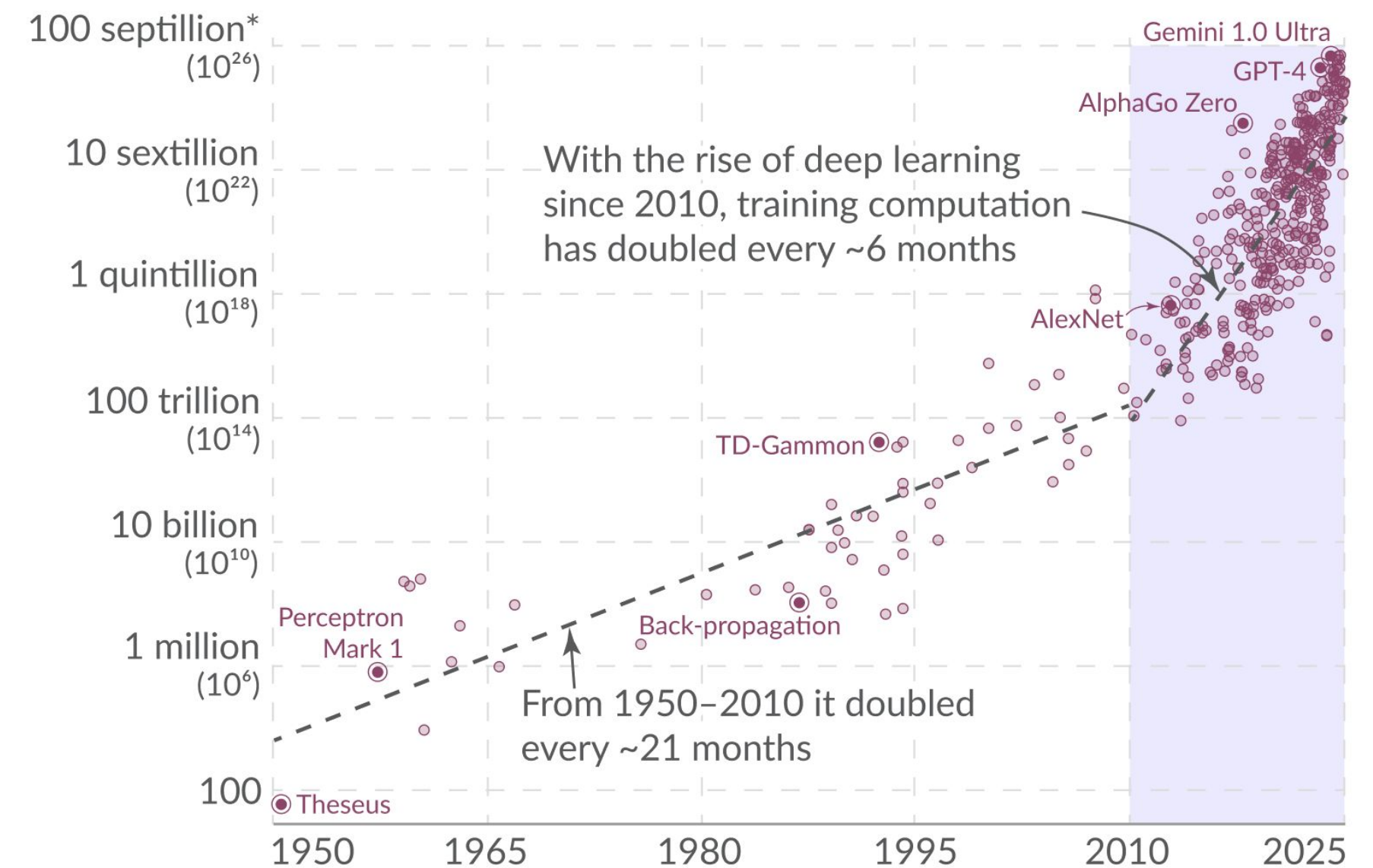
Why MLOps is needed?

- Resources for ML are growing
 - models training (gpu)
 - data mining (cpu, disks)
 - labeling (people)

The computation used to train notable AI systems has doubled every ~6 months since 2010

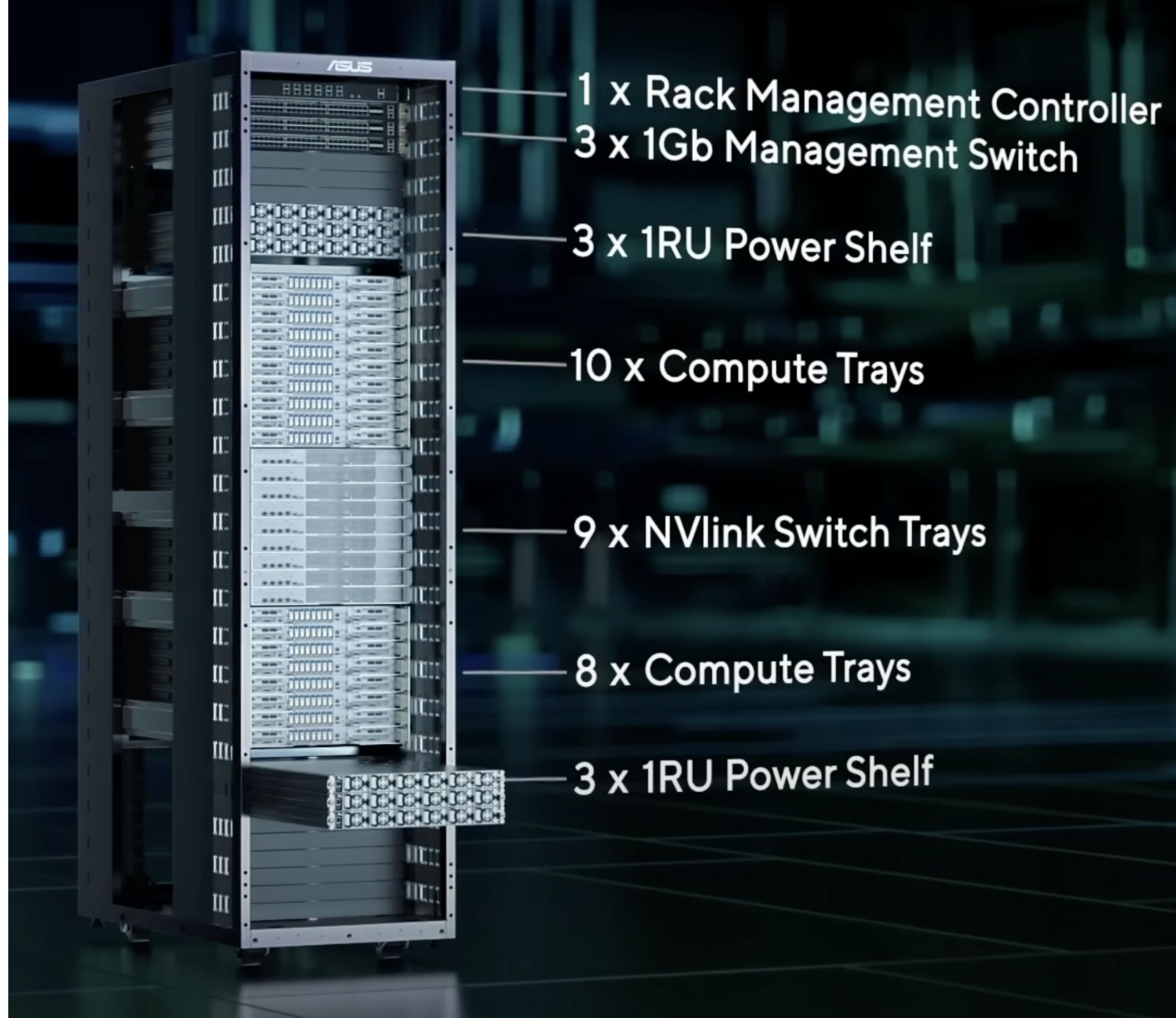
Our World
in Data

Training computation is measured in total floating-point operations (FLOP). Each FLOP represents a single arithmetic calculation, such as multiplication. Shown on a logarithmic scale.



*For comparison, 1 septillion (1,000,000,000,000,000,000,000,000) is the estimated number of stars in the universe.

Data source: Epoch (2024)



Why MLOps is needed?

- Resources for ML are growing
 - data mining (cpu)
 - labeling (people)
 - models training (gpu)
- Results reproduction
 - not only for industry, but for academia as well

Links: [one](#), [two](#), [three](#), [four](#), [five \[ru\]](#)

Why Most Published Research Findings Are False

🌐 1 language ▼

Article [Talk](#)

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From Wikipedia, the free encyclopedia

"**Why Most Published Research Findings Are False**" is a paper by Ioannidis et al. published in the [Stanford School of Medicine](#) journal *metascience*.

In the paper, Ioannidis argues that many published research findings contain results that cannot be reproduced. He determines whether scientific findings are based on formalized calculation ("P value") or on subjective assumptions about the results, which indicates that most published research findings are false.

COMMUNICATIONS OF THE ACM

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REVIEW ARTICLES

Threats of a Replication Crisis in Empirical Computer Science

By Andy Cockburn, Pierre Dragicevic, Lonni Besançon, Carl Gutwin

Communications of the ACM, August 2020, Vol. 63, No. 8, Pages 70-73

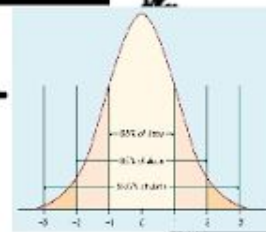
Why MLOps is needed?

- Resources for ML are growing
 - data mining (cpu)
 - labeling (people)
 - models training (gpu)
- Results reproduction
 - not only for industry, but for academia as well
- Models delivery
 - reduce time to market
 - reduce routine
 - testing and reliability
- Competence decomposition
 - deeper division of labor

Why do ML specialists need MLOps?

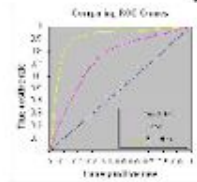
The Virgin Statistics

Entire skillset falls to pieces the moment they work with an uncommon distribution



Wears suits and cologne to seem more professional

Obsessed with plotting. Claims it helps people understand, but it only confuses them more.



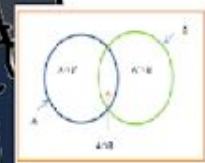
Convolved, ineffective methods get rebuffed at every work meeting.

"I k...know the results seem weird, b-but the p-value is considered significant..."

Decision is:	The Null Hypothesis is	
	True	False
Accept H_0	(1- α) Confidence Level	β
Reject H_0	α	(1- β) Power of the test

Checks his results.

Obsessed with theory. Can't do anything without stealing the methods of smarter, dead men.



Secretly resentful, wishes his field was more popular.

Constantly begging superiors for more data.

Knows *maybe* three methods, all requiring impossible assumptions.

Simple Linear Regression Model

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i$$

Labels: Dependent Variable, Possible Target, Regression Line, Independent Variable, Random Error

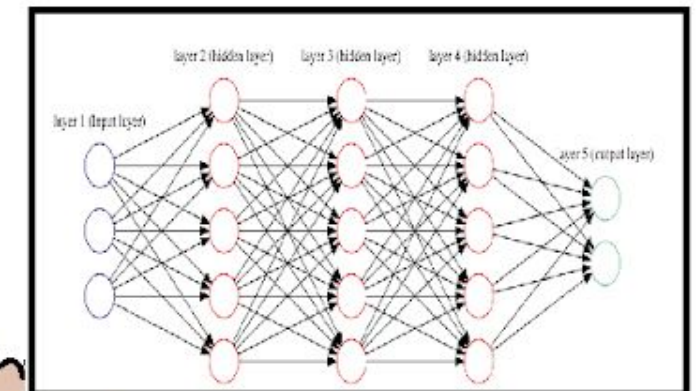
The Chad Machine-Learning

Regularizes his data until it gives the result he wants.



Has literally never heard the word "distribution" in his life

Fastest growing and most in demand discipline in his country.



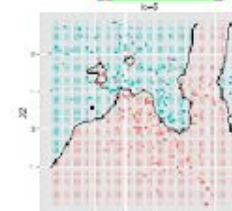
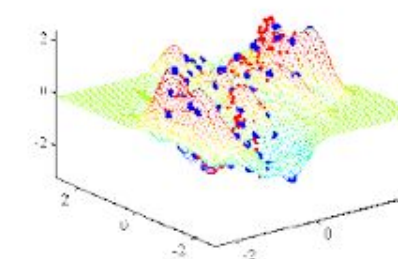
Innovates daily, advancing his field while solving real problems.

Doesn't try to explain methods, simply gives results, and is respected for it.

Wears sweats and muscle shirts to work, no one can stop him because he's too valuable.

Has so much data, often has to throw it away.

Let's his subordinates waste their days making his plots, would be a waste of his talent.



Refuses to use the same method more than once, each problem is unique and requires its own touch.

Implements methods into production without testing.

Why do ML specialists need MLOps?

because no one will do it for you!!!

For deeper dive I suggest reviewing
system design primer

<https://github.com/donnemartin/system-design-primer>

- cookbook for modern web based systems

Greg Brockman, OpenAI CTO

“... we need more people with **strong software skills who can deliver these systems**. This is because we are building AI models out of unprecedented amounts of compute...”

“Harnessing this compute **requires deep software skills** and the right kind of machine learning knowledge. We need to coordinate lots of computers, build software frameworks”

<https://blog.gregbrockman.com/its-time-to-become-an-ml-engineer>



What is MLOps?

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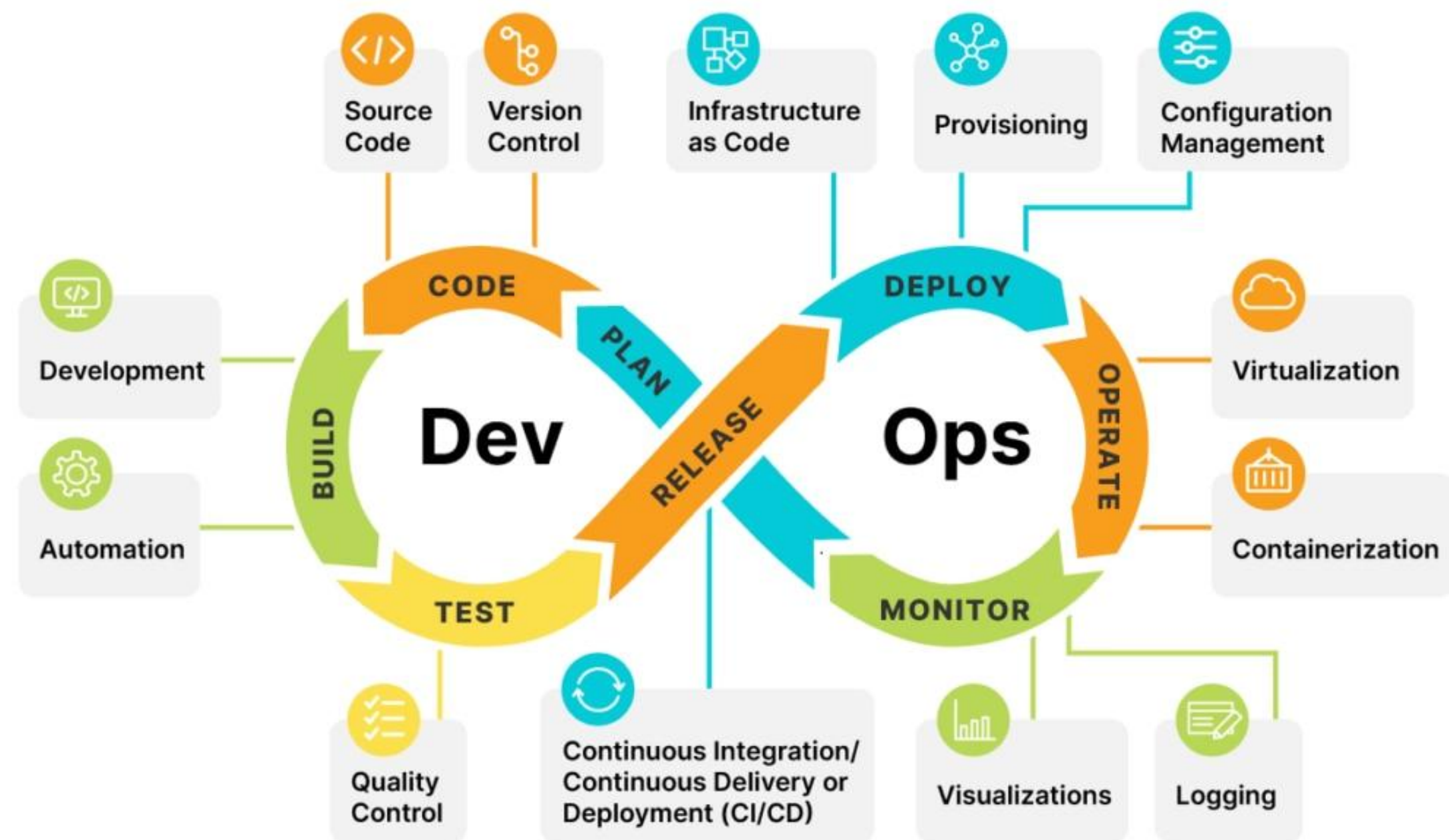
DevOps definition

DevOps is a methodology in the software development and IT industry. Used as a set of practices and tools.

DevOps integrates and automates the work of software development (Dev) and IT operations (Ops) as a means for improving and shortening the systems development life cycle

[Well-known common knowledge site](#)

[Great talk on DevOps practices](#)



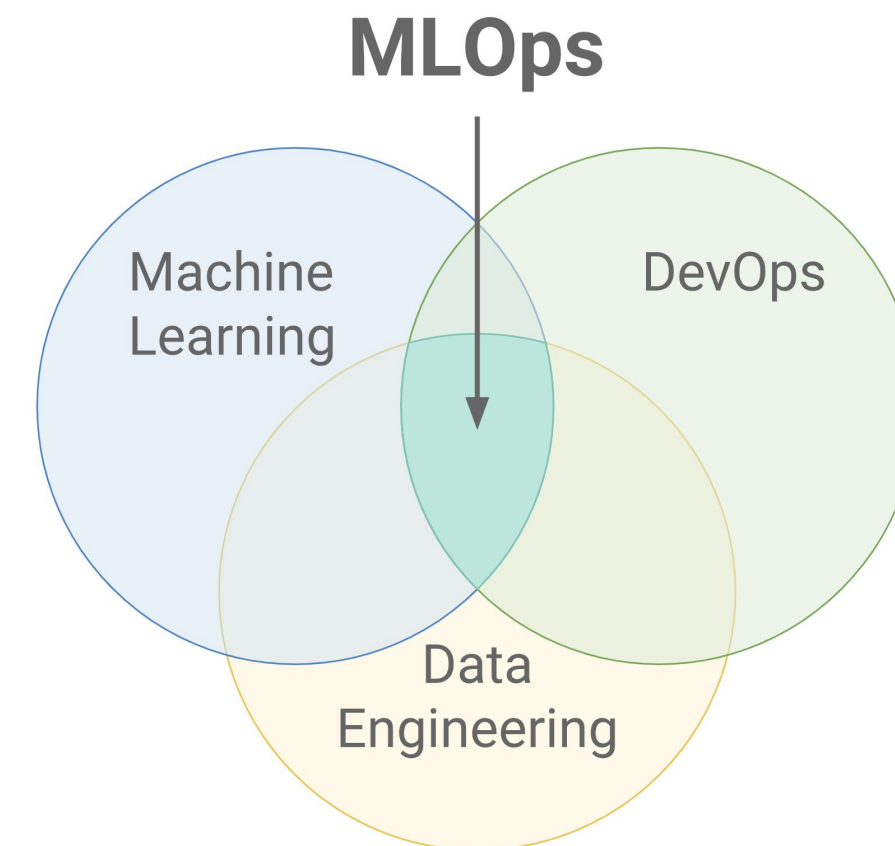
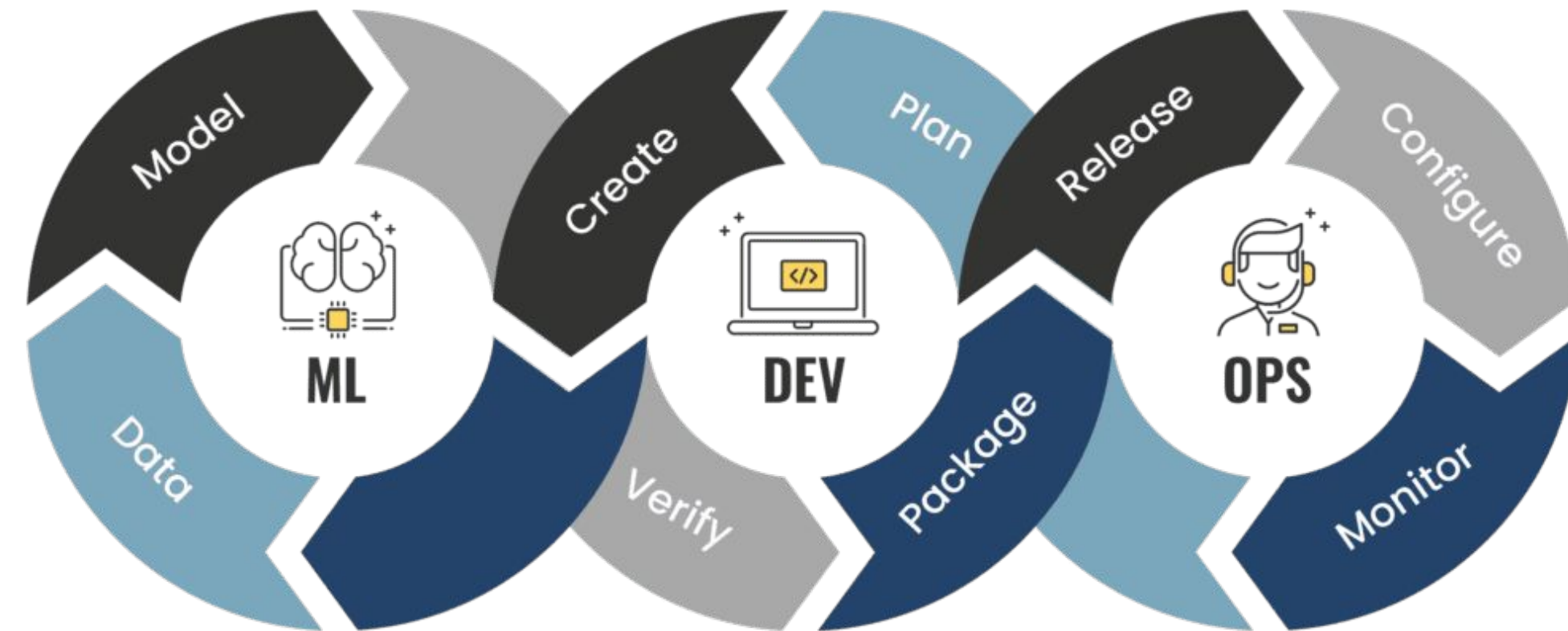
[DevOps course at MIPT \(ru\)](#)

MLOps definition

MLOps is a paradigm that aims to deploy and maintain machine learning models in production reliably and efficiently.

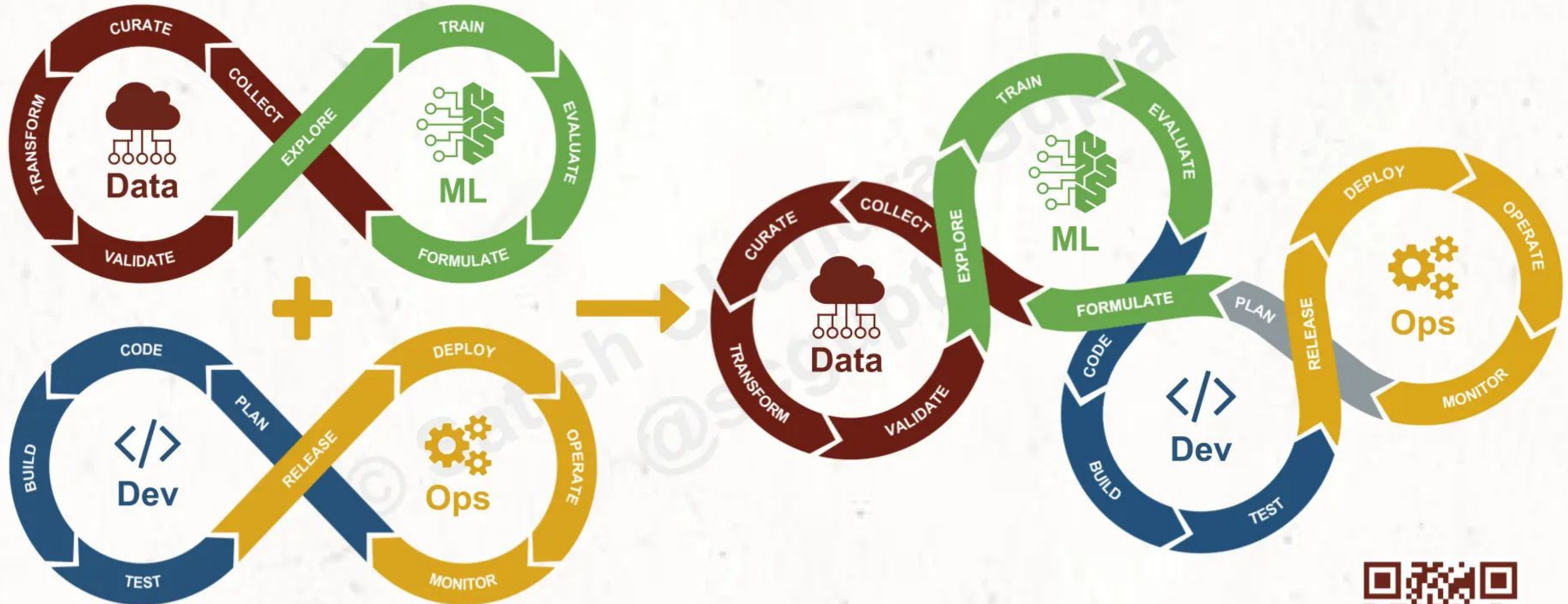
MLOps seeks to increase automation and improve the quality of production models, while also focusing on business and regulatory requirements

[Well-known common knowledge site](#)



MLOps = DataML + DevOps

ml4devs.com/mlops-lifecycle 



© 2022 Satish Chandra Gupta



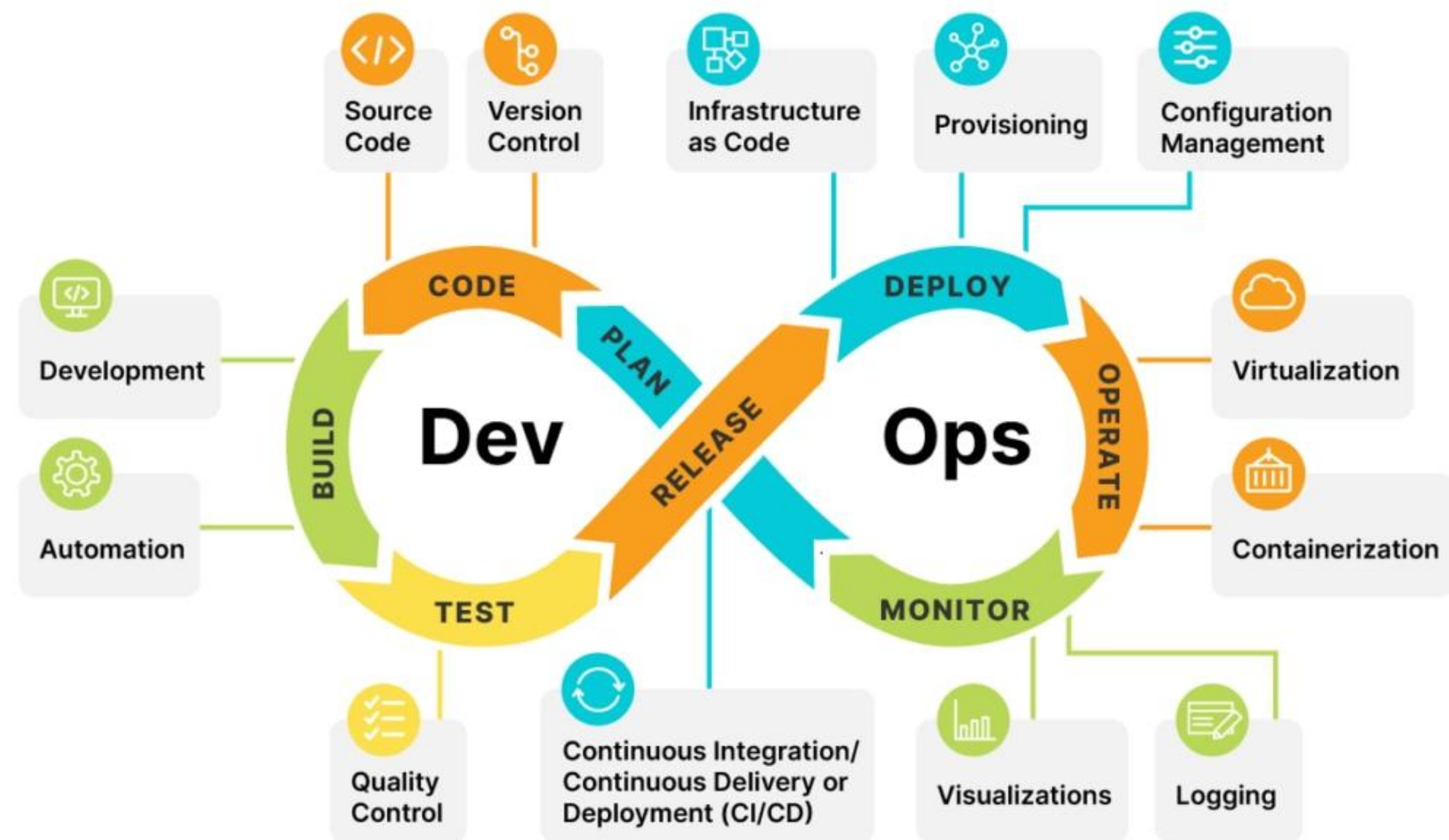
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twitter.com/scgupta 
linkedin.com/in/scgupta 



Relations to other spheres

- Management
- System Design
- Software development
- Data bases
 - SQL
- Analytics
- Hardware
 - Cluster design
- Machine Learning



Processes and management

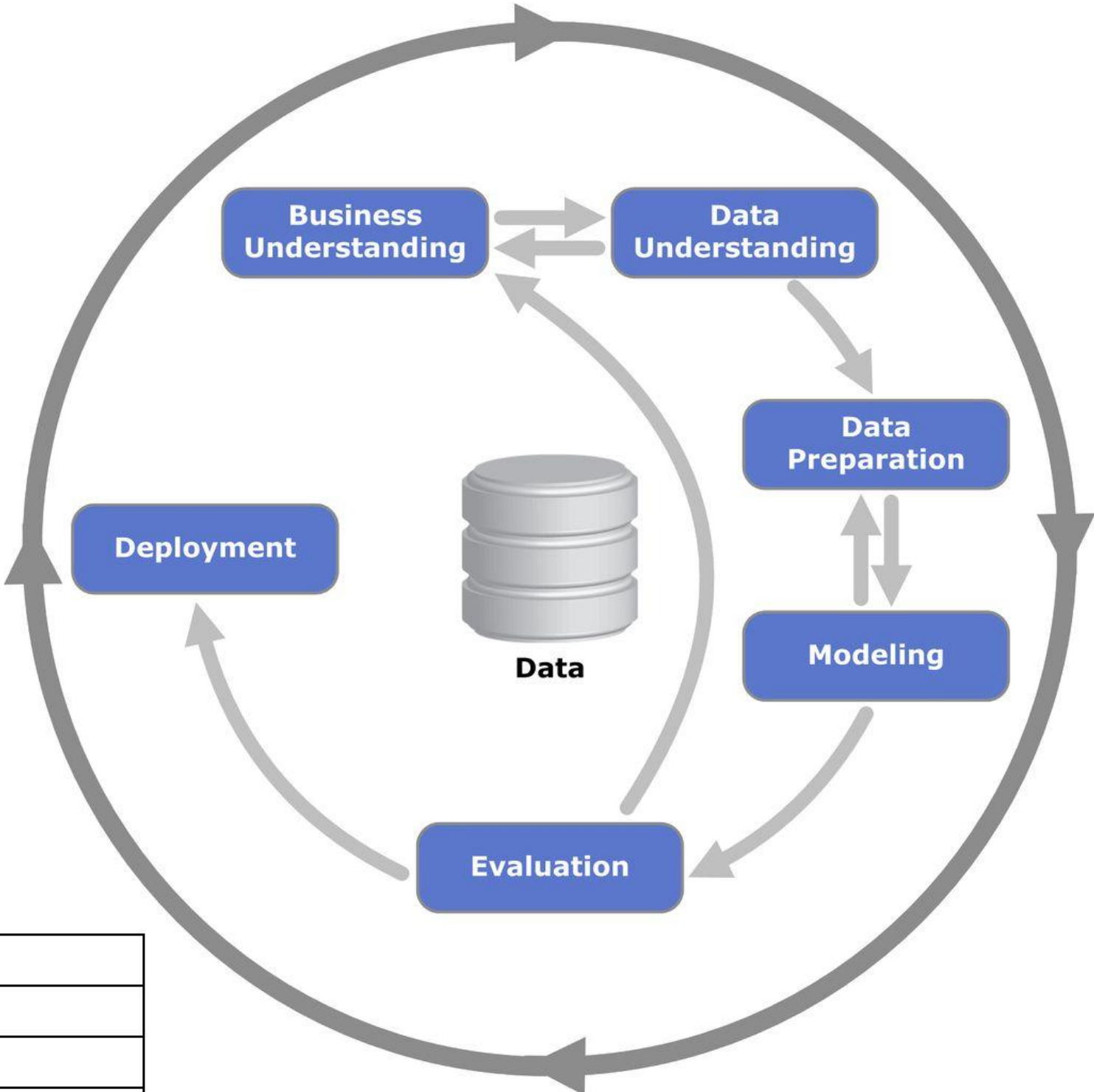
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CRISP DM

In the field of Data Science is used. **Cross-industry standard process for data mining** ([CRISP DM](#)) is a process standard for projects using predictive analytics, proposed in 1999. [It was used by half of the companies](#) engaged in Data Mining at the beginning of the century.

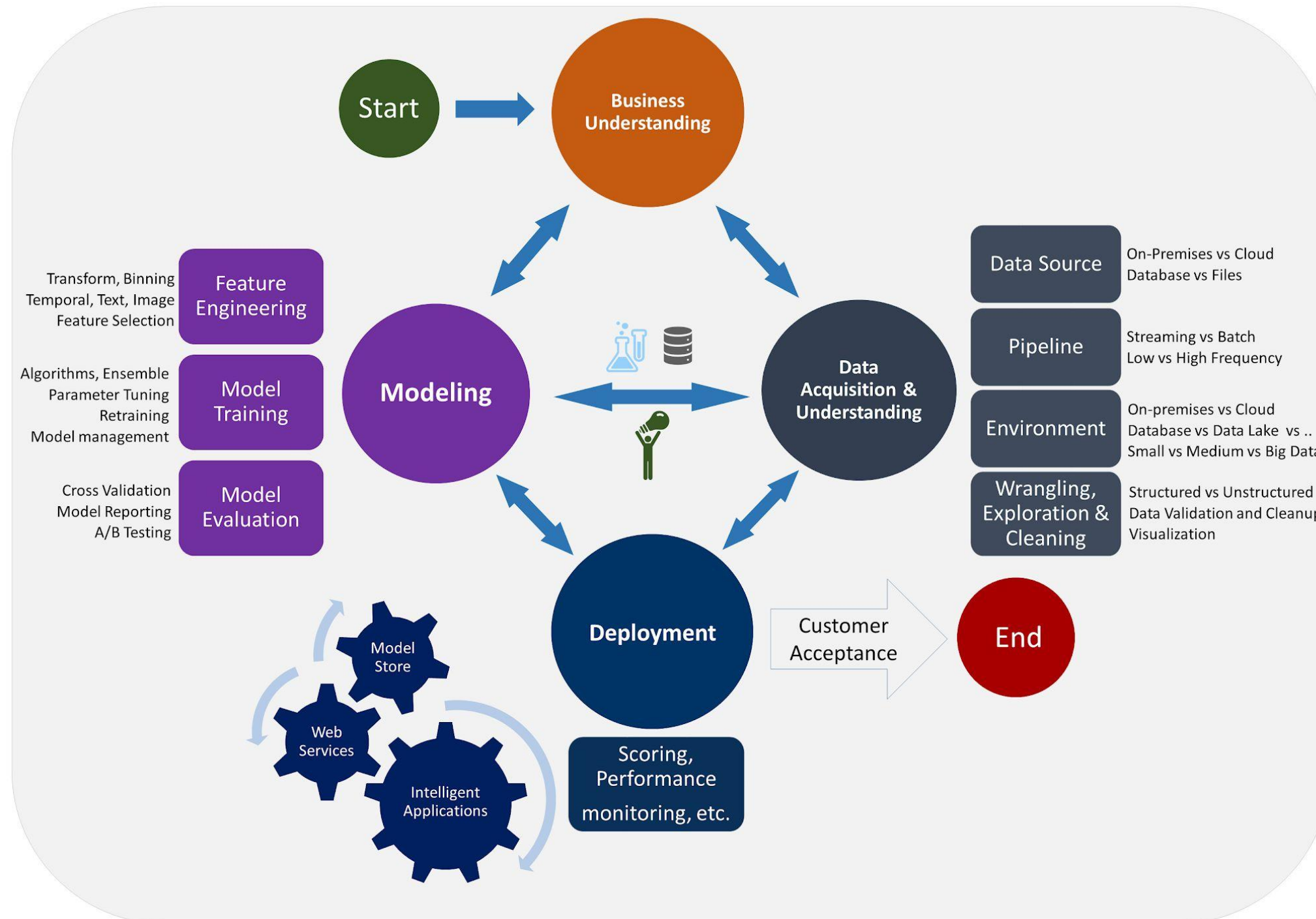
[Read more](#)



Poll Years	2002	2004	2007	2014
CRISP-DM	51%	42%	42%	43%
SEMMA	12%	10%	13%	8.5%
KDD process			7%	7.5%
My organization's	7%	6%	5%	3.5%
My own	23%	28%	19%	27.5%
Other (incl. domain specific)	4%	6%	9% (5%)	10% (2%)
None	4%	7%	5%	0%

Team Data Science Process

Data Science Lifecycle

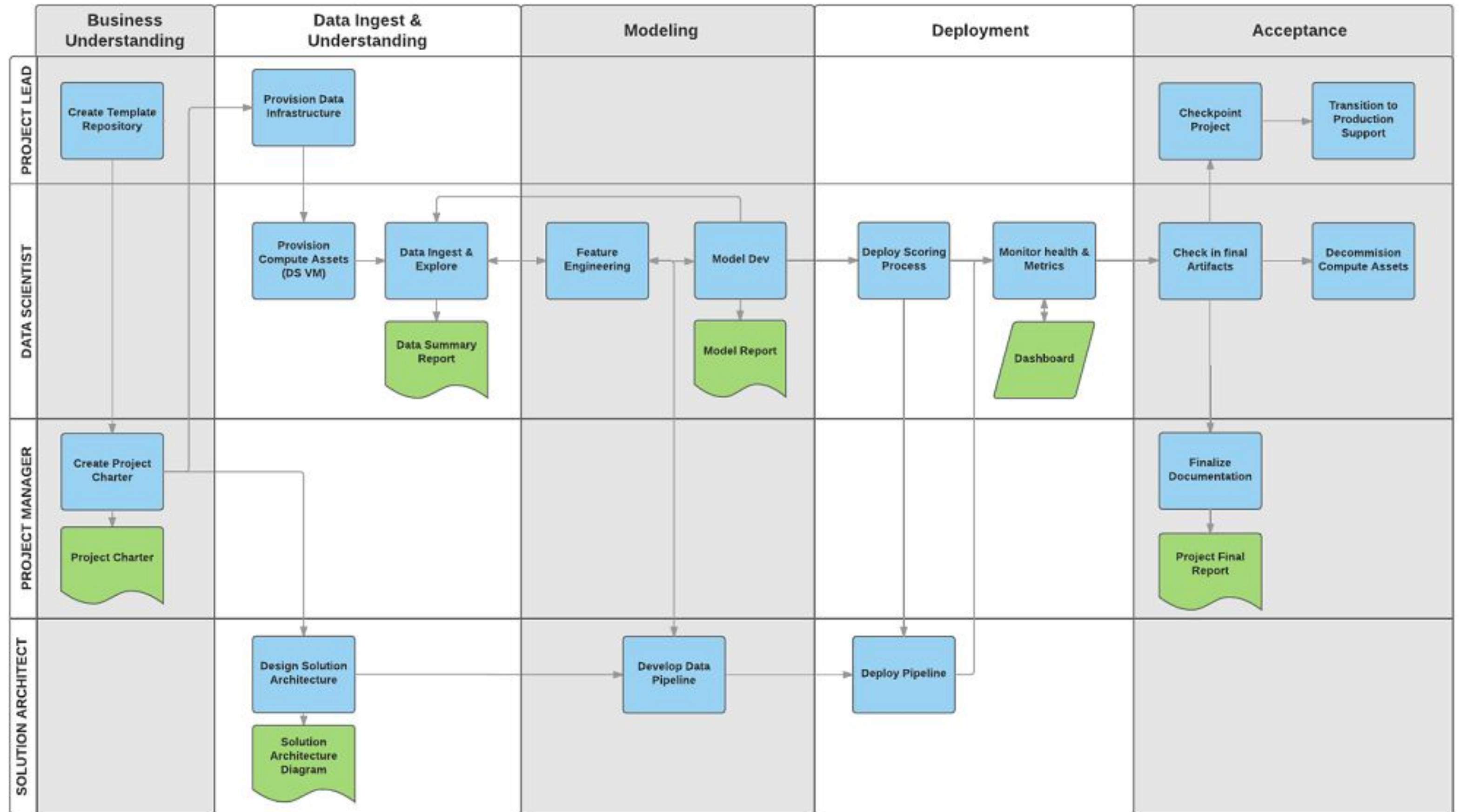


Modern standard from Microsoft

<https://learn.microsoft.com/en-us/azure/architecture/data-science-process/overview>

It is distinguished by the allocation of roles and a more detailed study of the stages of the cycle

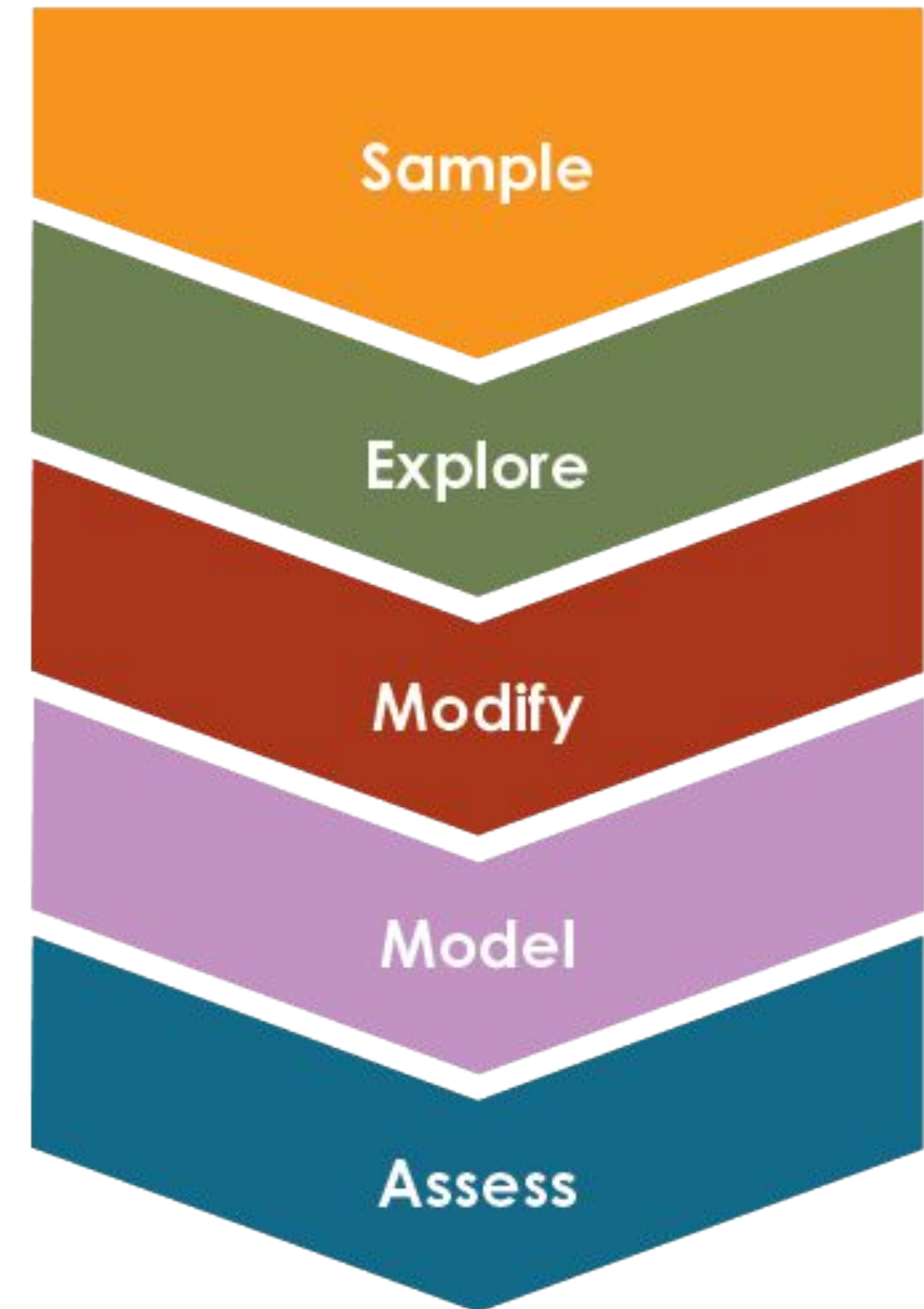
TDSP: team roles



Other approaches

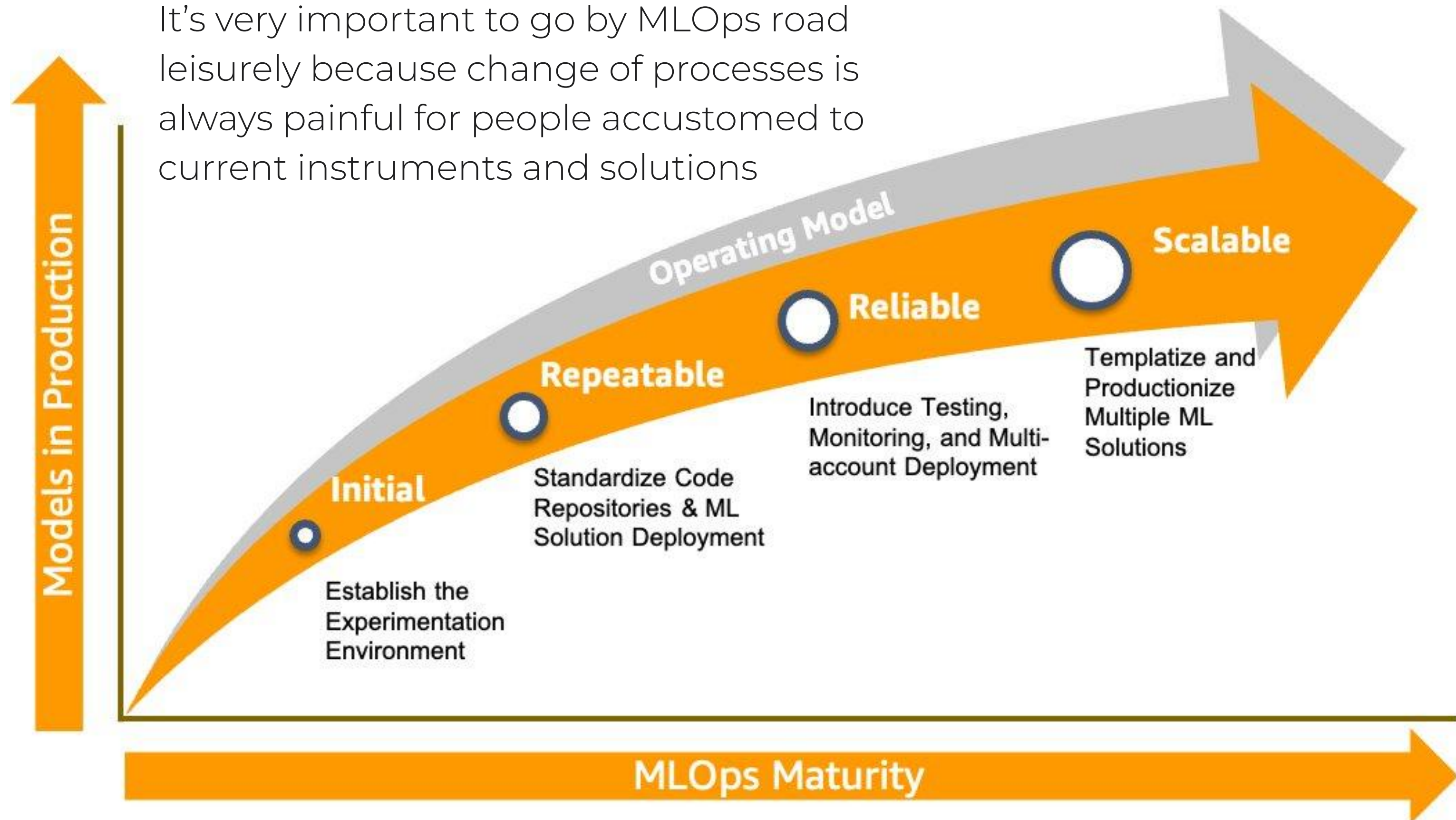
1. [SEMMA](#)
[read more](#)
2. [KDD Process](#)
3. [Waterfall](#)
4. etc...

In fact these are topics for IT management course



MLOps gradual introduction steps

It's very important to go by MLOps road leisurely because change of processes is always painful for people accustomed to current instruments and solutions



MLOps as a requirement for ML

<https://proceedings.neurips.cc/paper/2015/file/86df7dcfd896fc2674f757a2463eba-Paper.pdf>

Hidden Technical Debt in Machine Learning Systems

D. Sculley, Gary Holt, Daniel Golovin, Eugene Davydov, Todd Phillips
{dsculley, gholt, dgg, edavydov, toddphillips}@google.com
Google, Inc.

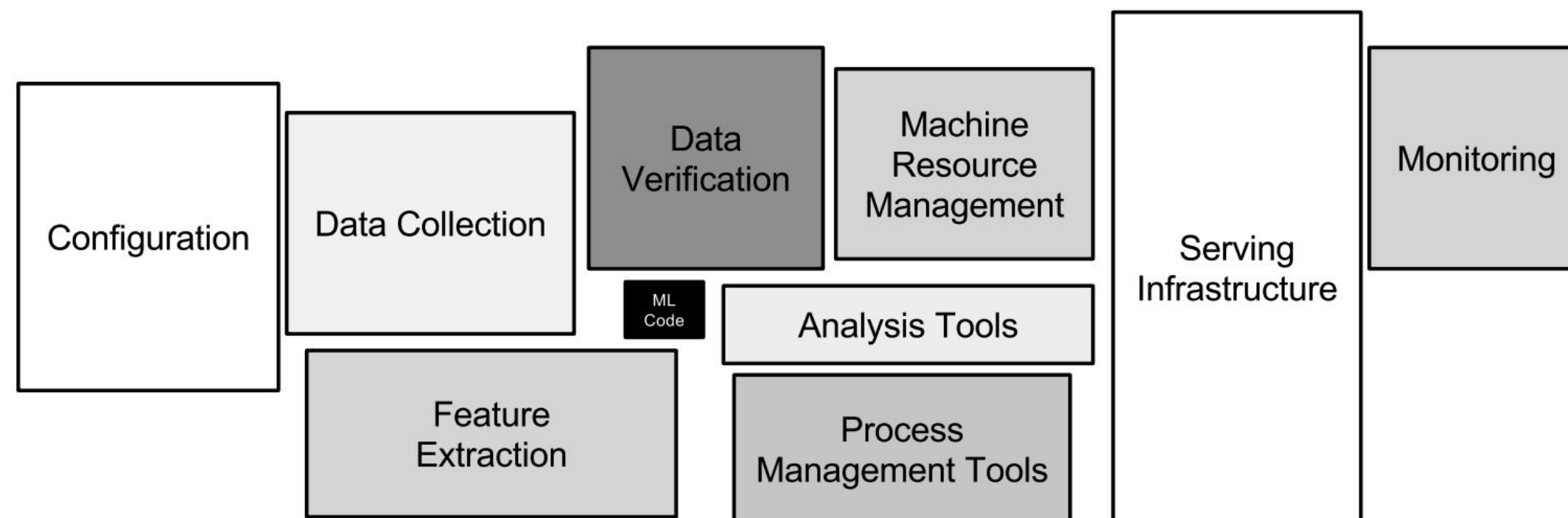
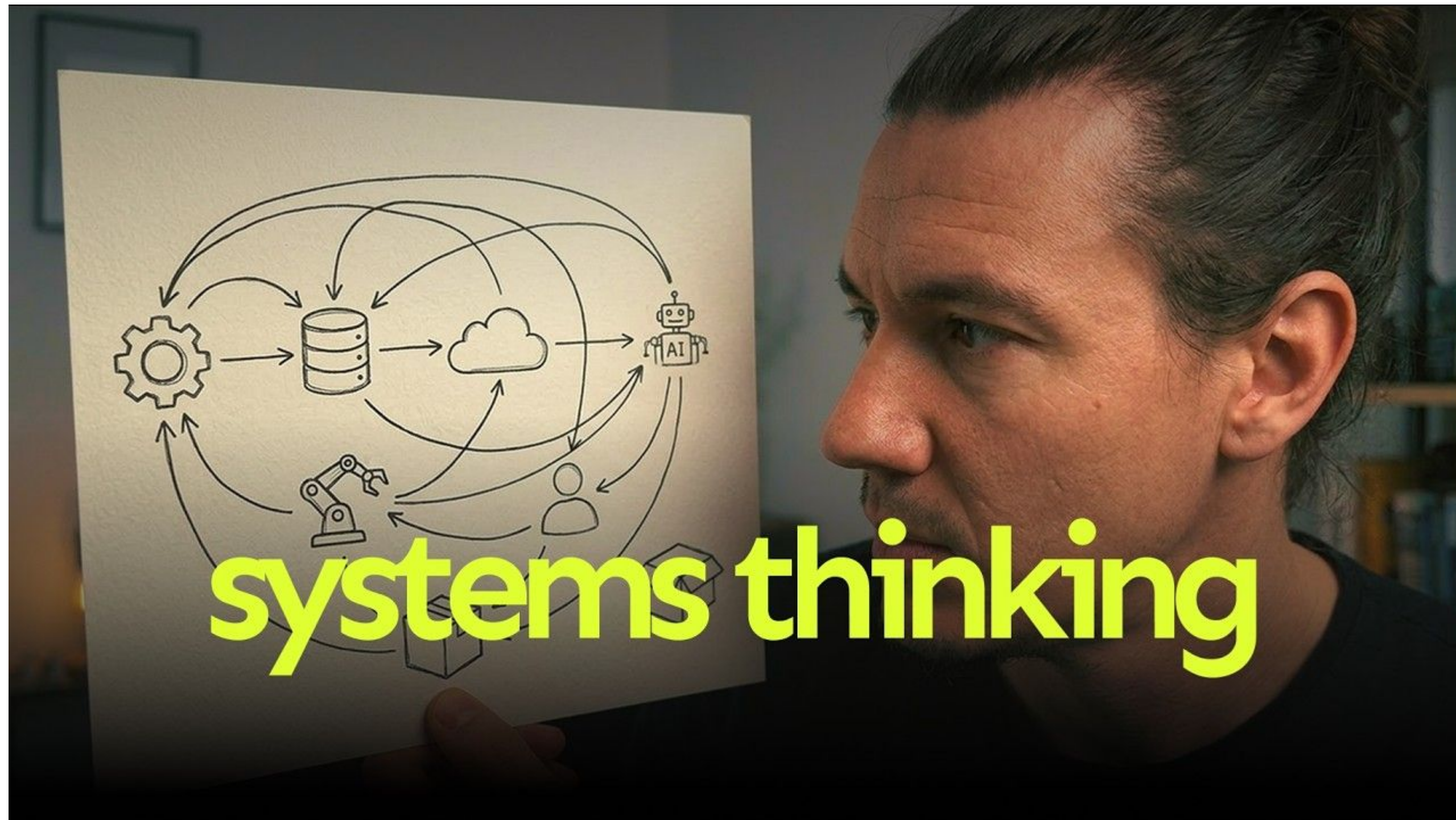


Figure 1: Only a small fraction of real-world ML systems is composed of the ML code, as shown by the small black box in the middle. The required surrounding infrastructure is vast and complex.

System thinking is the key

<https://www.youtube.com/watch?v=7zCsfe57tpU>



Further reading

- <https://ml-ops.org/content/mlops-principles>
- <https://www.ml4devs.com/en/articles/mlops-machine-learning-life-cycle/>
- <https://ashutoshtripathi.com/2021/08/18/mlops-a-complete-guide-to-machine-learning-operations-mlops-vs-devops/>
- <https://netraneupane.medium.com/introduction-to-mlops-for-beginner-part-1-37207ea3004a>
- <https://neptune.ai/blog/how-to-learn-mlops>

MLOps subtasks

Course structure

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MLOps subtasks

- Contemporary solutions overview
- Software engineering
 - Python infrastructure
- Virtualization
 - Docker
- Code storage and management
 - Git
 - CI & CD
 - code quality
 - testing (auto tests, test stands)
 - documentation generation
- Data storage and management
 - Data Acquisition
 - Data Engineering
 - AB tests

MLOps subtasks

- Contemporary solutions overview
- Software engineering
- Virtualization
- Code storage and management
- Data storage and management
- Labeling and crowdsourcing
- Experiments tracking
 - production monitoring
- Computational resources
 - GPU
- Model storages and deployment preparation
- Efficient inference
- Embedding storage optimization and vector databases
- LLMOps: vLLM, ollama and friends



MLOps *stacks* overview

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MLOps at big tech

Big companies have their stack to organize models training and publishing pipelines

- Bigtech
 - [Amazon SageMaker](#), [example](#)
 - [Google Cloud](#)
 - [Nvidia](#)
 - [Microsoft Azure](#)
 - Yandex: [YTSaurus](#) + Nirvana + [Toloka](#)
- Independent vendors
 - [Databricks](#)
 - [Seldon](#)

Generally variance of instruments is huge over different companies (from only bare metal ssh machines to very solid custom solutions)

Closed circle solutions

Pros:

- high level of integration
- reliability

Cons:

- high price
- binding to vendor
 - vendorlock



Not only big tech

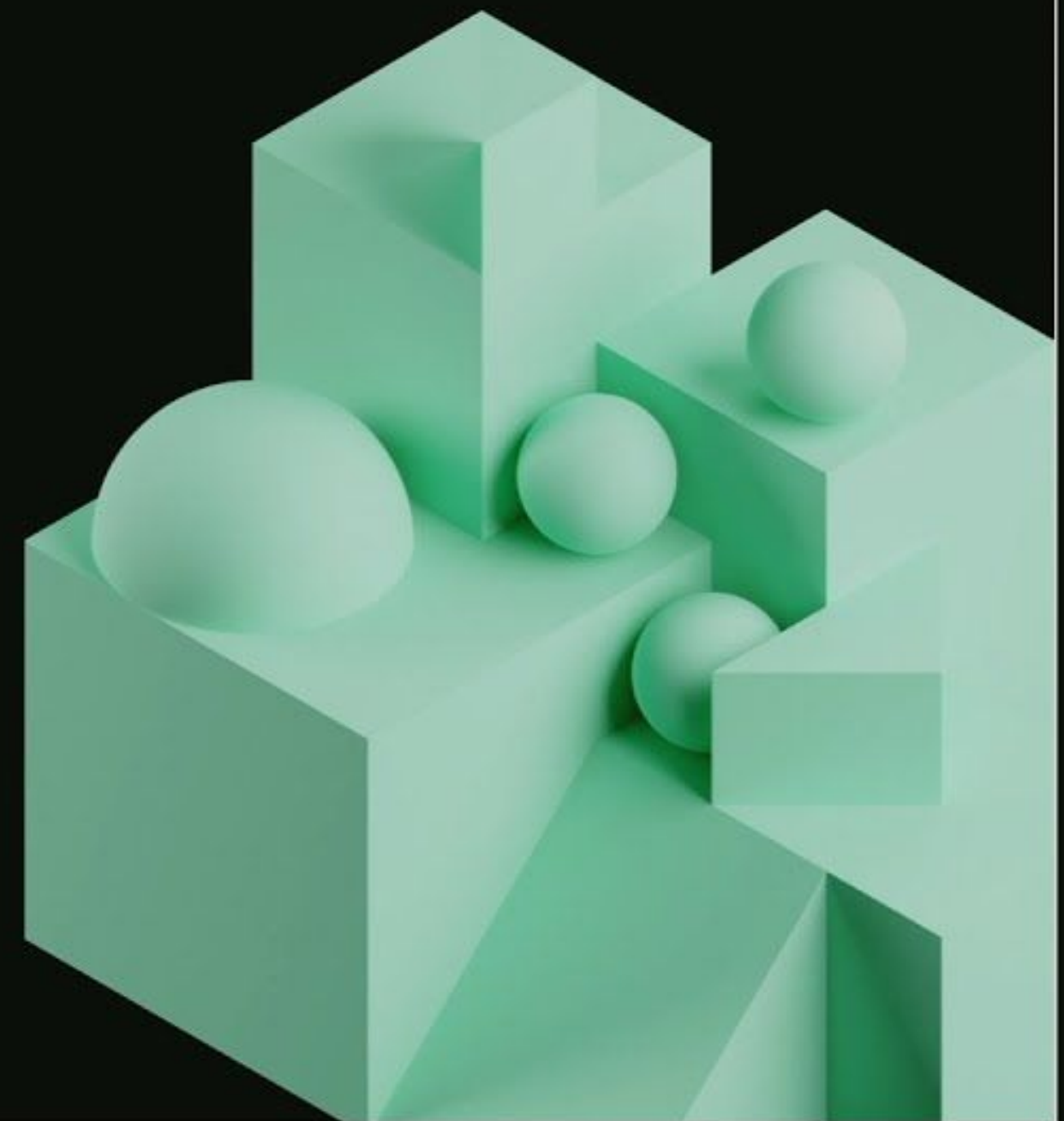
[Youtube video](#)

**DATA+AI
SUMMIT**
BY  databricks

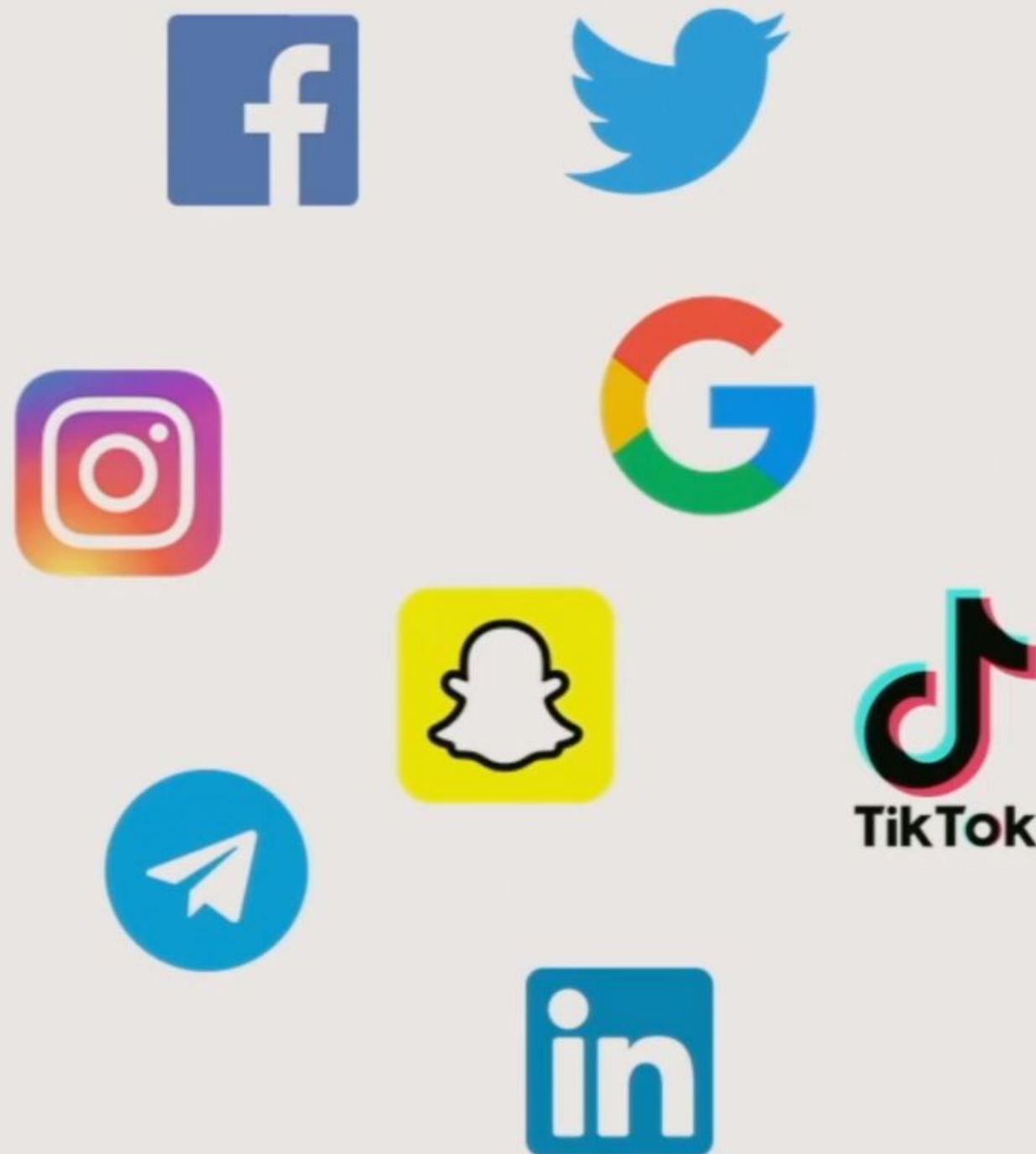
MLOps at Gucci: from zero to hero

An overview on implementing an MLOps
solution from scratch

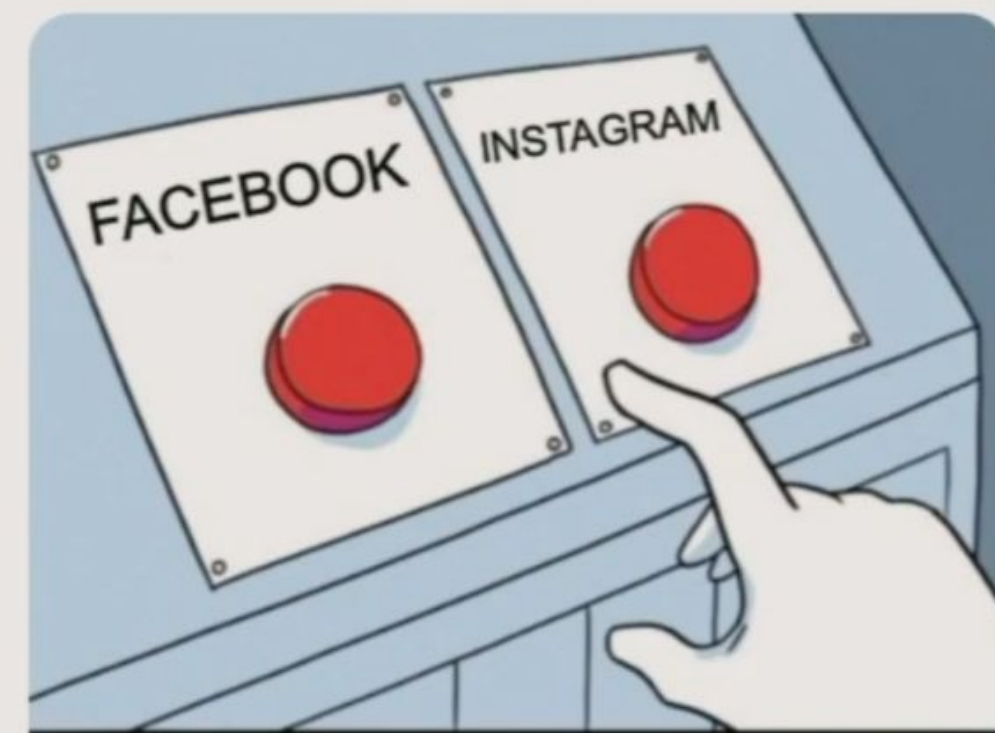

Databricks
2023



Paid digital media landscape is vast



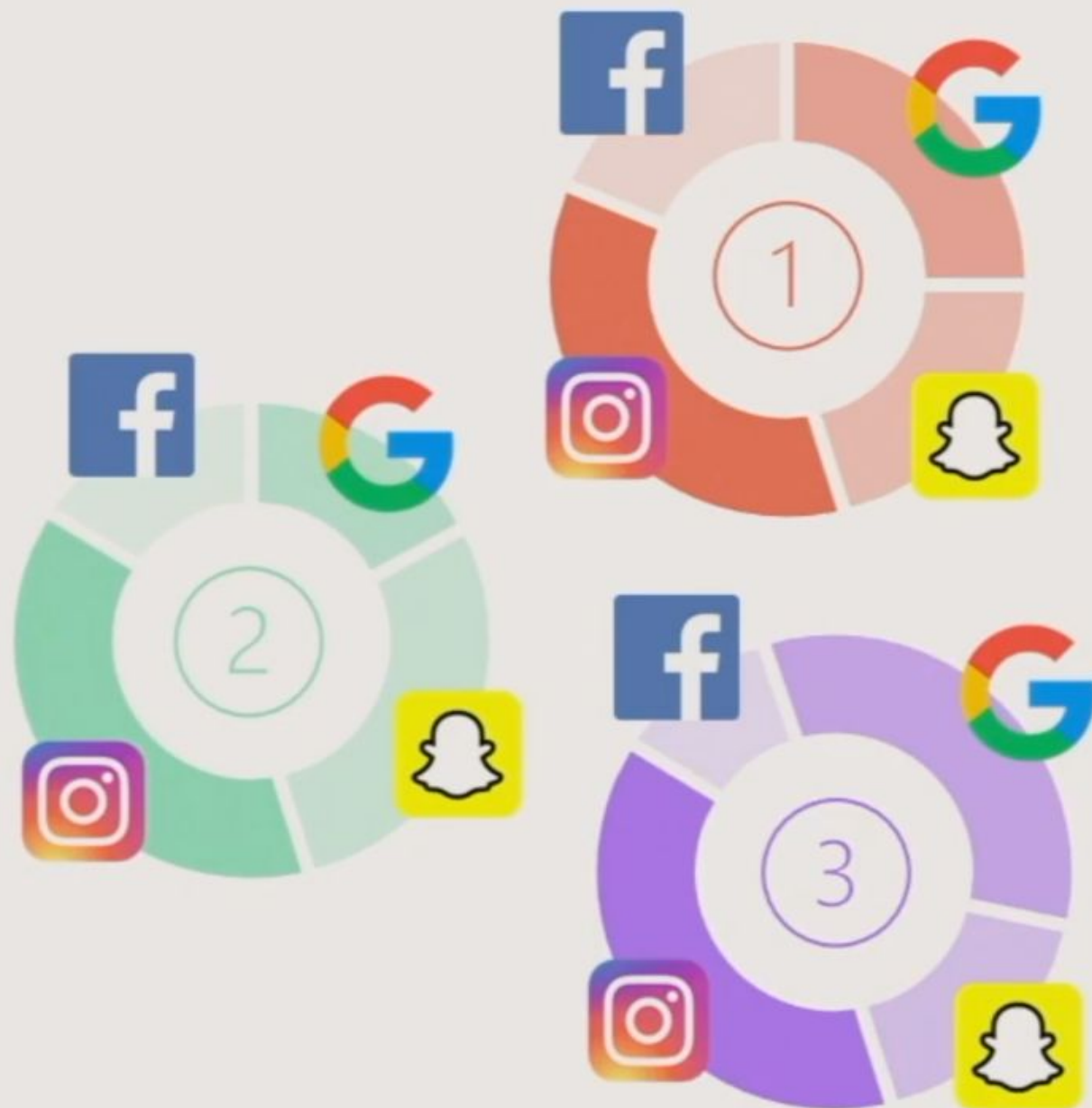
* террористы



Data driven decision

How to allocate budget on media channels to maximize ROI?

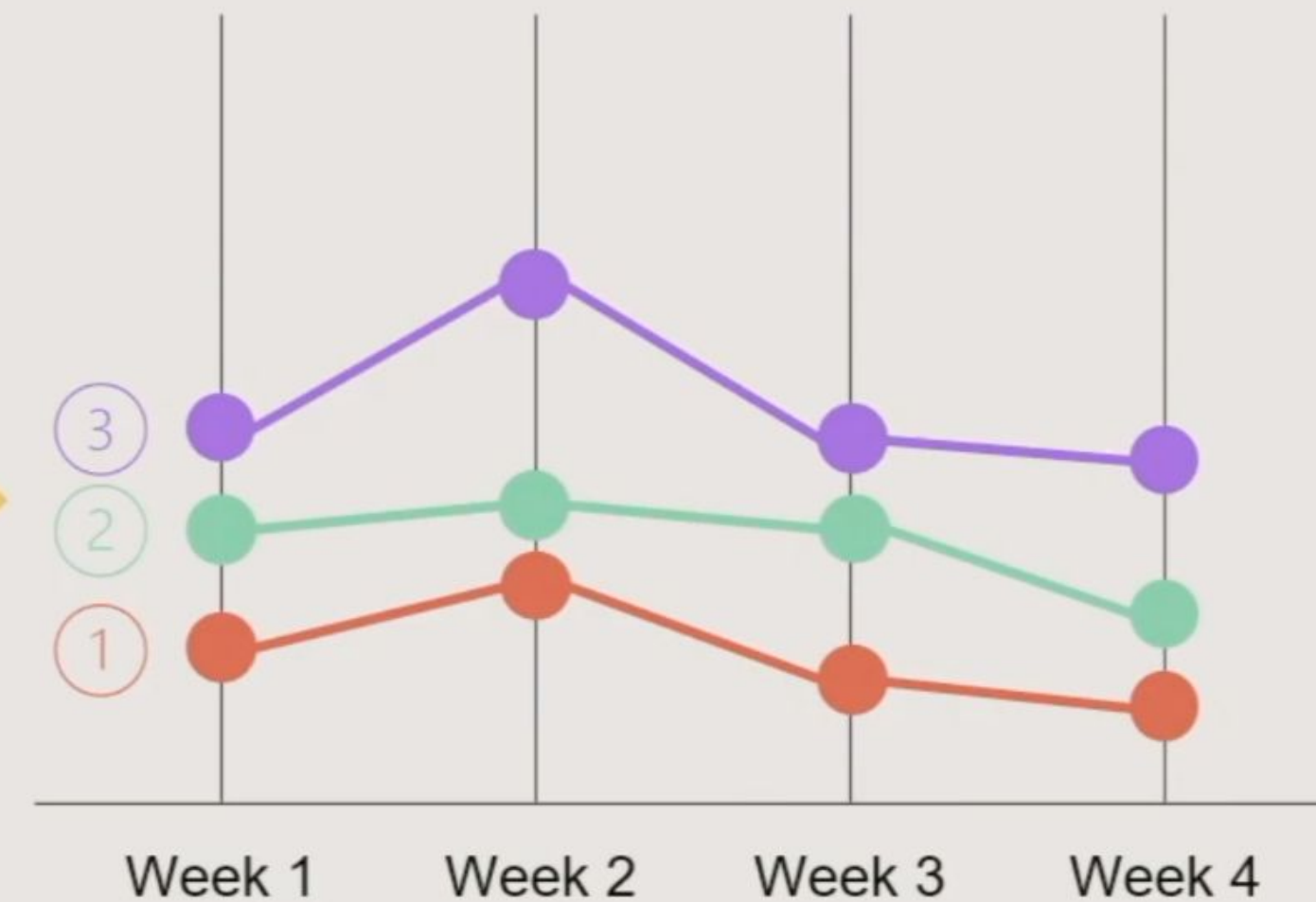
Budget scenarios



GROAS 2.0



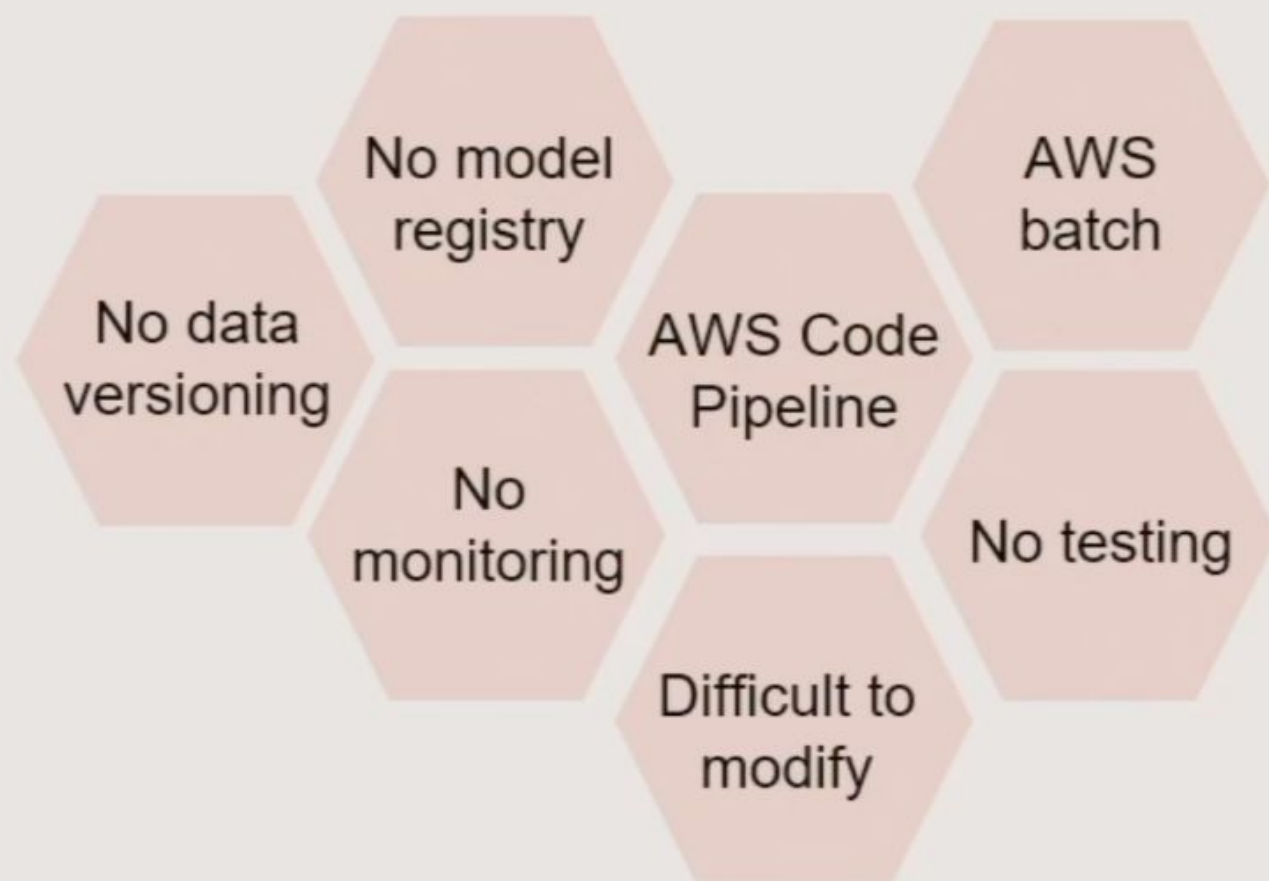
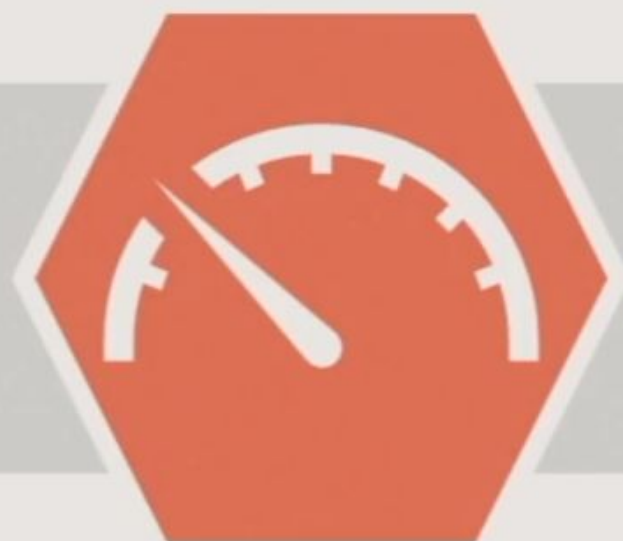
Net sales forecasts



Best scenario is chosen for future allocation

Get faster to production

GROAS 1.0
6 months (*)



GROAS 2.0
6 weeks (*)



MLOps at academia

It just doesn't exist (unfortunately)

You publish or perish:
no time for bullshit

E.g. GigaEvo



What about open source stack?

Will discuss it during all the course.

Main problem of FOSS instruments - **lack of orchestration**.

Each instrument primarily aims to solve one task but then start to add some default solutions for adjacent problems which are not that great.

All in all no instrument solve every problem.

Finally you need to **write glue code yourself**.

Agents

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Agents are everywhere...

Nowadays most of the practical code is written with agents rather than humans.

They also could be used to automate your workspace setup as well as remote server.

In this course we will use agents a lot in practical part.

Note: you still have to understand what's happening in your project.

Agents are not magic

Technically this is just a stack of backend technologies triggered by an LLM

Совет по культуре речи при Губернаторе Санкт-Петербурга.

ДАВАЙТЕ ГОВОРИТЬ КАК ПЕТЕРБУРЖЦЫ

Людмила ВЕРБИЦКАЯ —
один из авторов серии словарей
«Давайте говорить правильно»



371/A3/12

Вместо модных слов из AI-буллшита
можно использовать русские:

Скилл	- папка с markdown-файлом
Skill marketplace	- репозиторий с md-файлами
Хук	- скрипт, прописанный в JSON
Агент	- <code>while(tokensSpent < limitPer5h)</code>
Память агента	- файл на диске
Оркестрация агентов	- Promise.all
Планировщик	- markdown-файл с галочками
Self-healing	- retry с текстом ошибки в промпте
Рефлексия	- дозапрос к модели с "ты уверен?"
Guardrails	- if

«Слово — одежда всех фактов, всех мыслей»
М. Горький

Совместный проект Санкт-Петербургского метрополитена
и телеграм-канала t.me/GroksCareer

Skills are fundamental

Agent Skills are a lightweight, open format for extending AI agent capabilities with specialized knowledge and workflows.

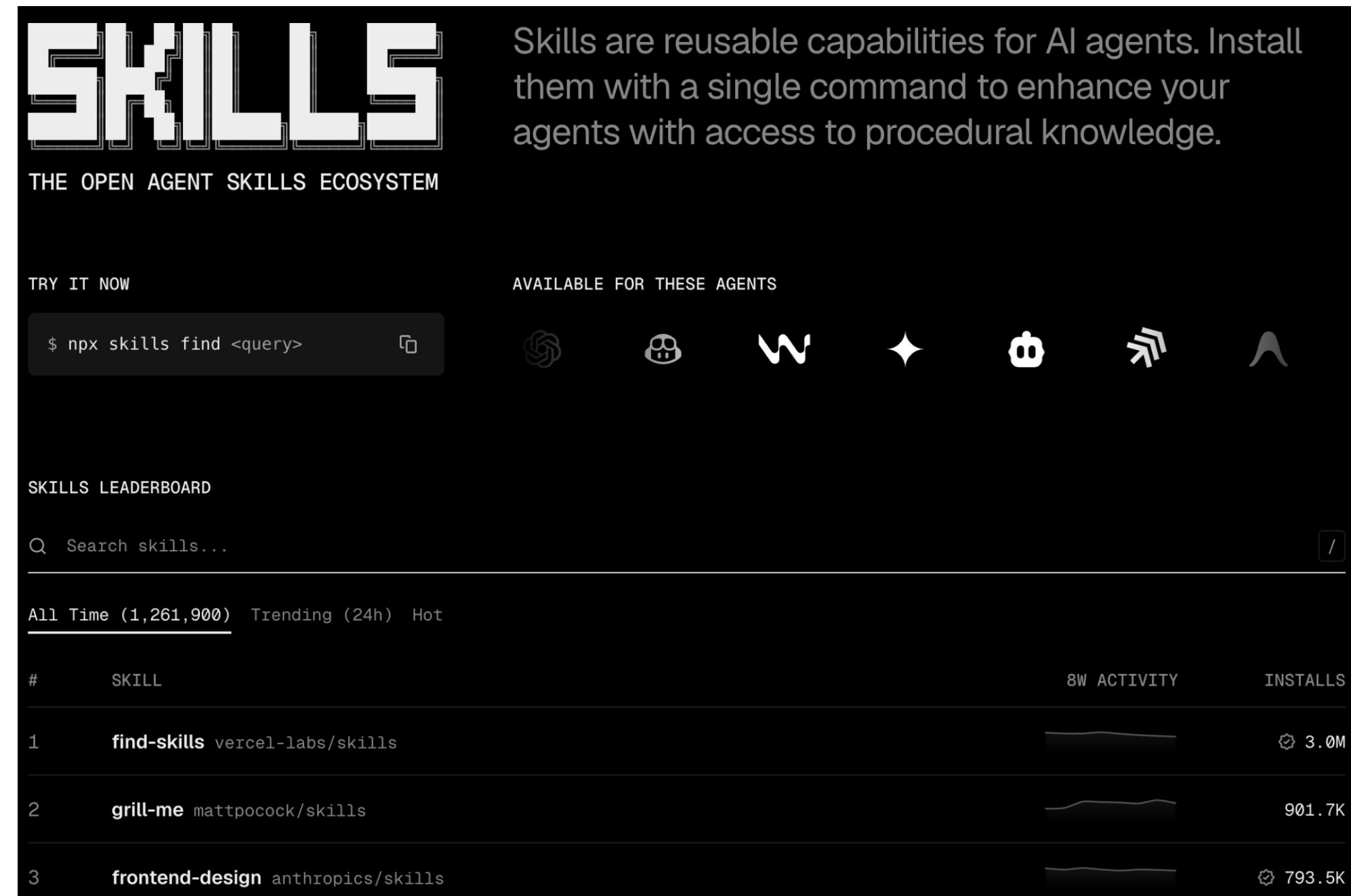
At its core, a skill is a folder containing a SKILL.md file. This file includes metadata (name and description, at minimum) and instructions that tell an agent how to perform a specific task. Skills can also bundle scripts, reference materials, templates, and other resources.

<https://agentskills.io/home>, <https://arxiv.org/html/2607.25032v1>

```
my-skill/
├── SKILL.md           # Required: metadata + instructions
├── scripts/          # Optional: executable code
├── references/        # Optional: documentation
├── assets/            # Optional: templates, resources
└── ...                # Any additional files or directories
```

Ways to find skills

- just ask an agent to look for skills needed
 - with **/find-skills** skill
- browse skill library yourself <https://www.skills.sh/>



The screenshot shows the Skills website interface. At the top, the word "SKILLS" is displayed in a large, stylized font, followed by the tagline "THE OPEN AGENT SKILLS ECOSYSTEM". Below this, there's a section "TRY IT NOW" with a command prompt input field containing "\$ npx skills find <query>". To the right, under "AVAILABLE FOR THESE AGENTS", several AI agent logos are shown, including OpenAI, Anthropic, and others. The main section is the "SKILLS LEADERBOARD", which has a search bar and tabs for "All Time (1,261,900)", "Trending (24h)", and "Hot". The "All Time" tab is selected, showing a table of skills ranked by popularity.

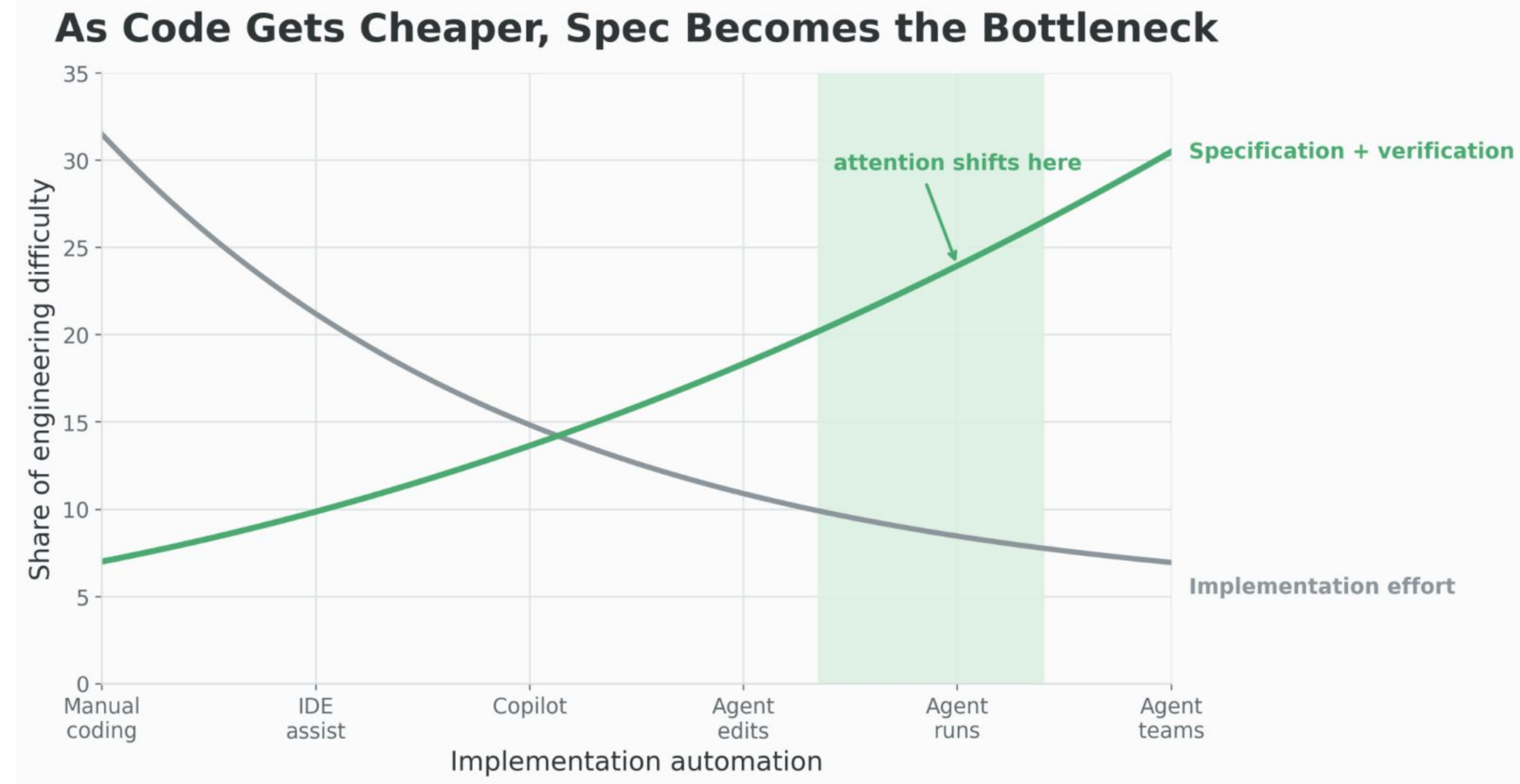
#	SKILL	8W ACTIVITY	INSTALLS
1	find-skills vercel-labs/skills		3.0M
2	grill-me mattpocock/skills		901.7K
3	frontend-design anthropics/skills		793.5K

How to code with agents?

The Right Amount of Spec for Agentic Development

When code gets cheap, the hard part is deciding what “correct” means and building a reliable way to check it.

ADR (architectural decision records) - very powerful technique to design your system in an agent first manner.

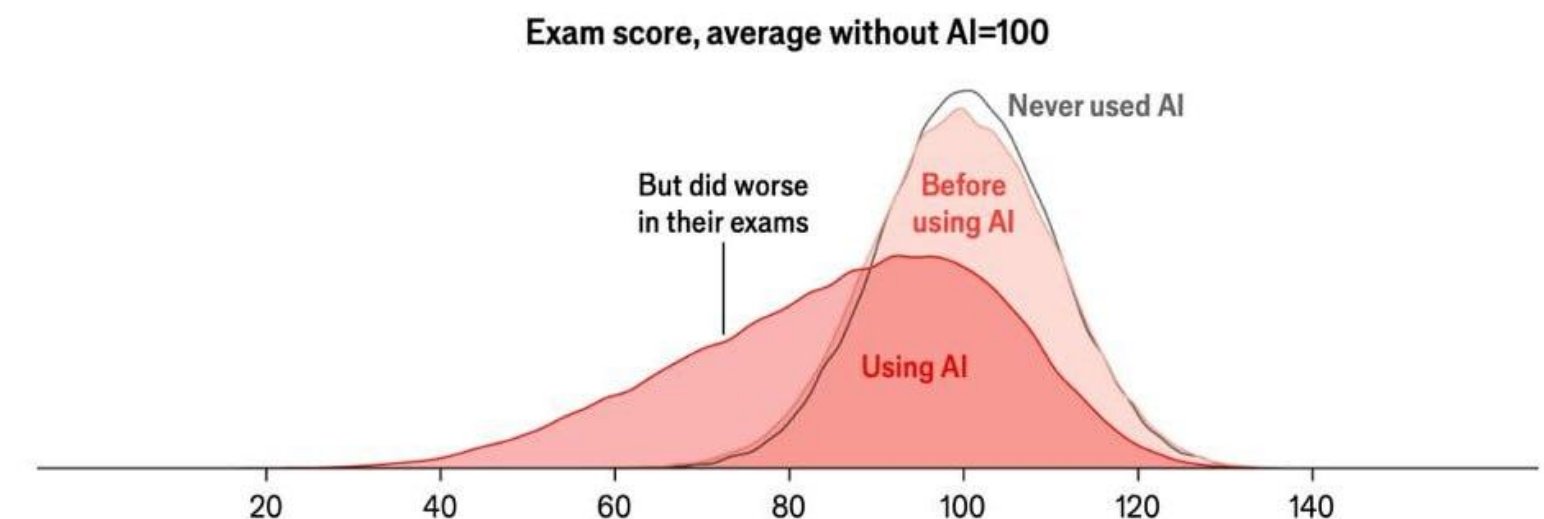
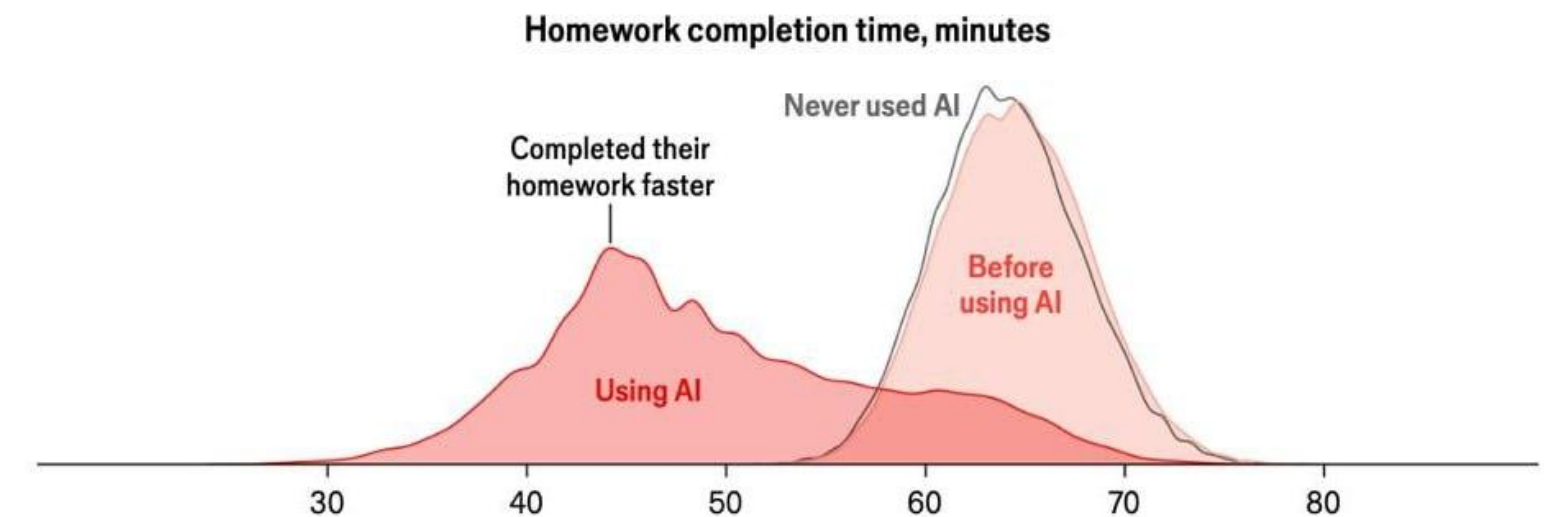
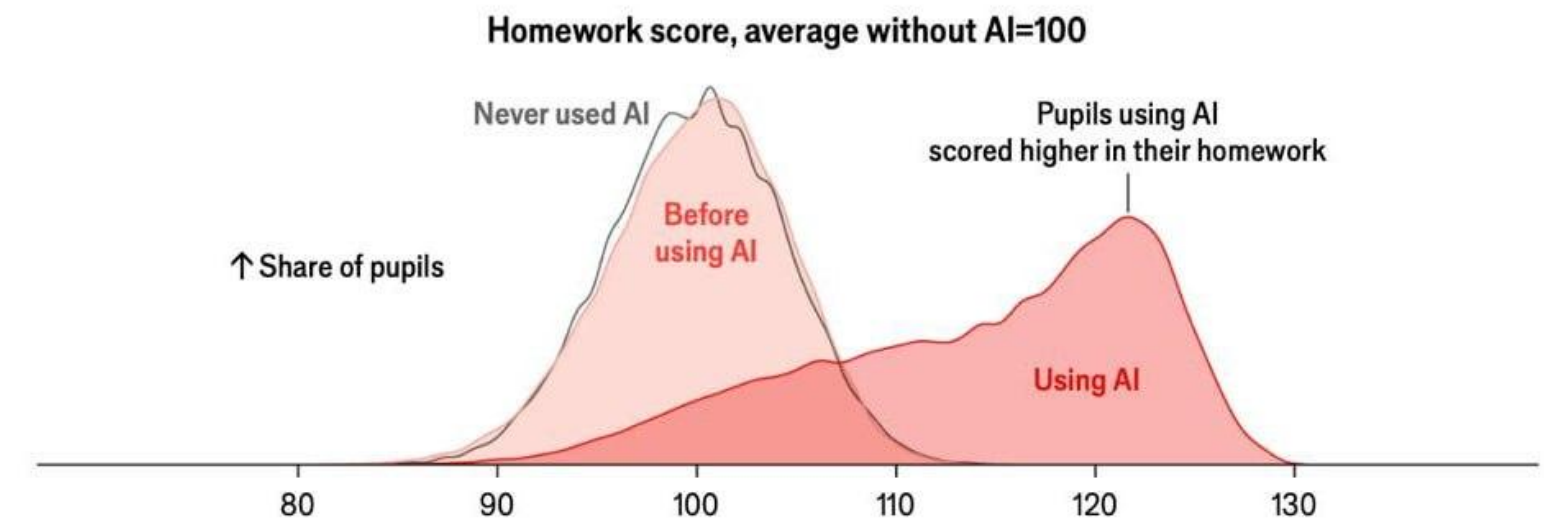
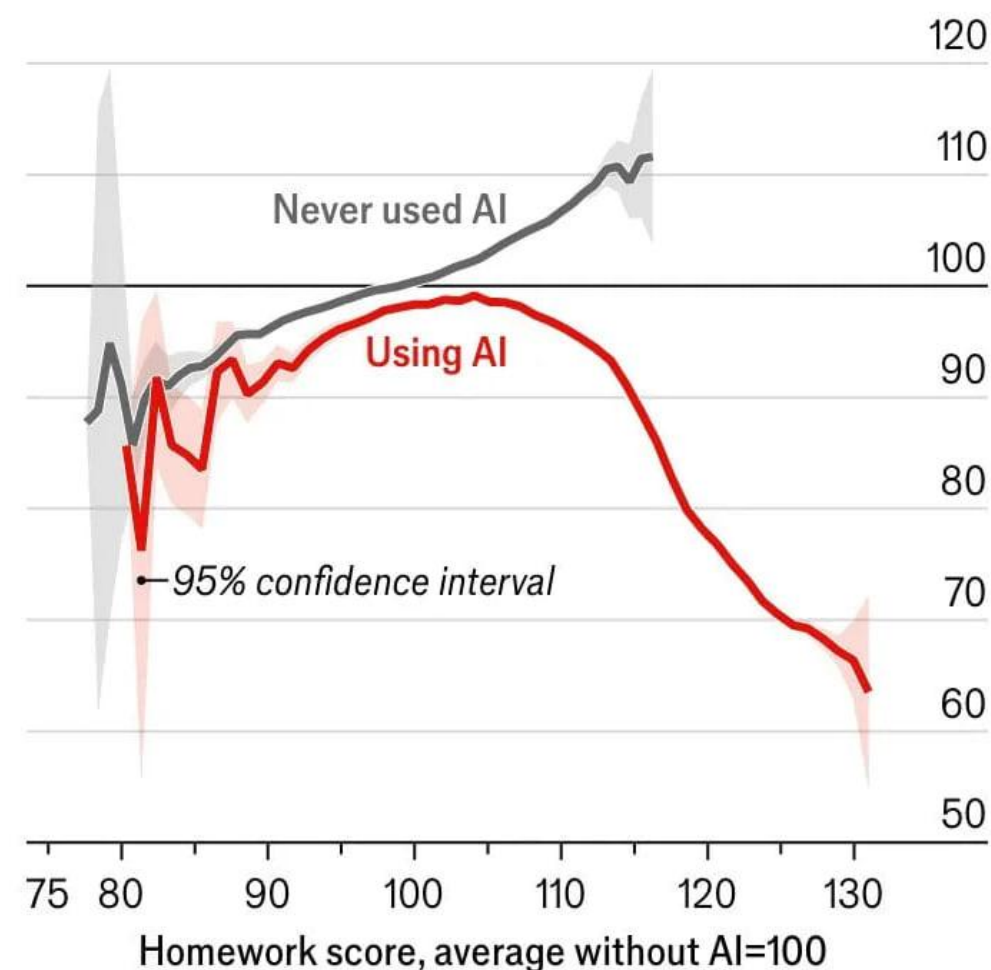


Learning with agents

- <https://www.skills.sh/mattpocock/skills/teach>
- <https://t.me/vgcourse/564>

Thinking about it

China*, average exam score
Jan 2023-Jun 2025, average without AI=100



Source: "The generative AI learning penalty: evidence from Chinese secondary education", by D. Strömberg et al., preprint June 2026

*26,811 pupils aged 12-18

Homeworks

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Homeworks

1. Project Proposal
 - a. Formulate problem to apply MLOps stack
2. Write baseline code
3. Apply parts of pipeline from classes
4. Peer review
5. Final review

Also we will have simple tests for each lecture.

One person does one project, no grouping in this course.

Bonus task: create pull request with some mlops instrument to any significant open source project e.g. [python-dotenv](https://github.com/python-dotenv/dotenv) lacks contemporary dependency management tool

Resources on MLOps

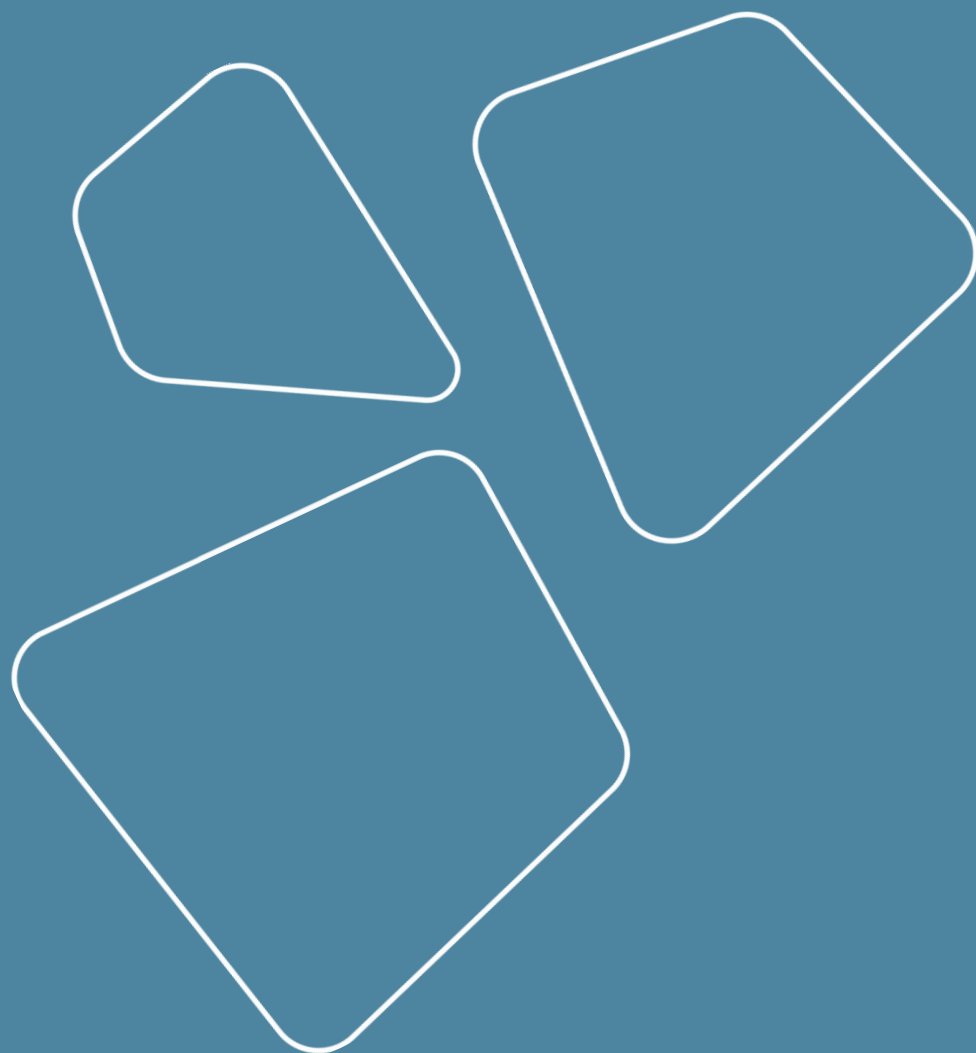
- <https://github.com/DataTalksClub/mlops-zoomcamp>
- <https://github.com/mryab/efficient-dl-systems/tree/main>
- <https://mlops-coding-course.fmind.dev/>
-

MLOps books

- Practical MLOps: Operationalizing Machine Learning Models by Noah Gift, Alfredo Deza
- Designing Machine Learning Systems: An Iterative Process for Production-Ready Applications by Chip Huyen
- AI Engineering: Building Applications with Foundation Models by Chip Huyen
-
-

Revise

1. Prerequisites
2. Motivation
3. What is MLOps?
4. Subtasks
 - a. and course structure
5. Stacks overview
6. Homeworks outline



Thanks for attention!

Questions?

